



LIST KERJAAN MINGGUAN

Timeline Proyek Sistem GLD

20 April 2026 – 7 Juni 2026 | 7 Minggu

M1

20–26 Apr



Desain Chamber



Uji Node GLD lama



Uji routing CH_Hello &
CH_Config

M2

27 Apr–3 Mei



Design.md (Algoritma
Sistem)



Uji Node GLD baru



Design Easy EDA
Chamber



Uji Konsumsi baterai CH

M3

4–10 Mei



Uji fungsional sistem
(GLD-CH-CW-Server)



Pembuatan Chamber &
Uji Sensor



Uji Daya panel surya ke
CH

M4

11–17 Mei



Pengujian Sensor + ML



Uji jarak & reabilitas
Komunikasi data



Uji Konsumsi baterai
Node GLD

M5

18–24 Mei



Uji Konsumsi baterai
Node GLD



Uji sistem di
lingkungan kampus

M6

25–31 Mei



Uji sistem lingkungan
kampus



Evaluasi

M7

1–7 Jun



Persiapan pemasangan
sistem



Selesai



In progress

Kerjaan Minggu ke-1

01 Desain Chamber

✓ Done

02 Uji Node GLD lama

✓ Done

03 Uji routing CH_Hello dan CH_Config

✓ Done

Pengujian Board Sensor

```
--- I2C Scan (direct bus) ---  
0x71 found  
Total: 1 device(s)
```

```
--- I2C Scan per TCA9548 channel ---  
ch0: 0x60  
ch1: 0x60  
ch2: 0x60  
ch3: 0x60  
ch4: 0x60  
ch5: 0x60  
ch6: 0x60  
ch7: 0x60
```

```
--- ADS1256 All Channels ---  
AIN0: raw= 2796248 1.666694 V  
AIN1: raw= 3600133 2.145846 V  
AIN2: raw= 2934370 1.749021 V  
AIN3: raw= 2568856 1.531157 V  
AIN4: raw= 2408928 1.435833 V  
AIN5: raw= 2363616 1.408825 V  
AIN6: raw= 2872853 1.712354 V  
AIN7: raw= 2942468 1.753848 V
```

```
[5] MCP4725 @ 0x60... ch0: MCP4725 FOUND  
ch1: MCP4725 FOUND  
ch2: MCP4725 FOUND  
ch3: MCP4725 FOUND  
ch4: MCP4725 FOUND  
ch5: MCP4725 FOUND  
ch6: MCP4725 FOUND  
ch7: MCP4725 FOUND
```

```
[6] ADS1256 SPI... 10ms  
ADS1256 SPI OK 10ms
```

PSU 5V 10A -> Circuit 5V Board (5.05V)

Tegangan VCC ESP32 = 3.305V

VSYS : 2,573V

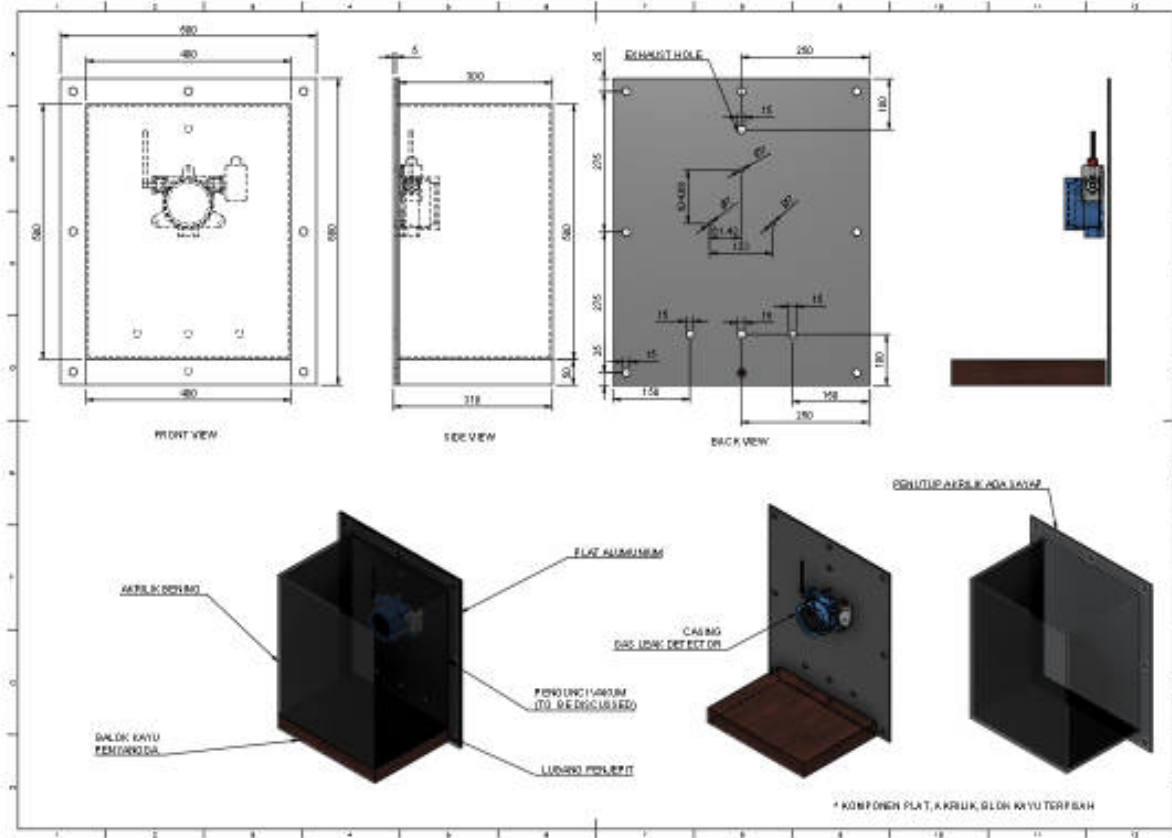
VBUS : 4.97V

```
--- I2C Scan per TCA9548 channel ---  
ch0: 0x60  
ch1: 0x60  
ch2: 0x60 0x62  
ch3: 0x60  
ch4: 0x60  
ch5: 0x60  
ch6: 0x60  
ch7: 0x60
```

```
--- I2C Scan per TCA9548 channel ---  
ch0: 0x60  
ch1: 0x60  
ch2: (empty)  
ch3: 0x60  
ch4: 0x60  
ch5: 0x60  
ch6: 0x60  
ch7: 0x60
```

```
Konek ke WiFi 'Fshares'Disconnected (GetOverlappedResult failed (PermissionError(13, 'Access is denied.', None, 5)))
```

Desain Chamber Training Gas



Cluster Head

Gas Leak Detector System

Skenario 1

Jaringan Normal

Proses inisialisasi & pembentukan topologi mesh



STEP 1

CH3 / GATEWAY Aktif

Address:	CH3 = 0x0003
U1 RX:	921 MHz (discovery + CH1 direct)
U3 RX:	922 MHz (CH2 data)
WiFi:	Connecting... → GAGAL → Retry
MQTT:	OFFLINE → OK ✓
GW Siap:	cluster=0x0003
HB:	rx=0 ok=0 WiFi=OK MQTT=OK

```
16:02:57.144 ->
16:02:57.144 -> === CH3 / GATEWAY (0x0003) ===
16:02:57.144 -> U1@921MHz: discovery+CH1direct | U3@922MHz: data CH2
16:02:57.144 -> !!! NYALAKAN INI PERTAMA !!!
16:02:57.144 ->
16:02:57.144 -> [WiFi] aqms-modem-001.....
16:03:12.199 -> [WiFi] GAGAL
16:03:12.199 -> [U1-921] Init... OK
16:03:12.284 -> [U3-922] Init... OK
16:03:12.318 -> [U1] Listening 921MHz (discovery + CH1 direct)
16:03:12.318 -> [U3] Listening 922MHz (CH2 data)
16:03:13.221 ->
16:03:13.221 -> [GW] Siap | cluster=0x0003 | MQTT:OFFLINE
16:03:13.221 ->
16:03:15.783 -> [MQTT] OK
16:03:24.933 -> [HB|30s] rx=0 ok=0 WiFi=OK MQTT=OK
```

STEP 2

CH2 Aktif & Melakukan Config

Address: CH2 = 0x0002

U1 RX: 921 MHz (dari CH1)

U3 TX: 922 MHz (ke CH3)

Discovery: Broadcast 1/5 @ 922MHz

Response: 0x0003 RSSI=-112.0dBm (2x)

Parent: 0x0003 @ -112.0dBm

CH_HELLO: → 0x0003 → Terkirim ✓

Status: CH2 Siap | parent=0x0003

```
16:04:48.733 -> === CH2 (0x0002) ===
16:04:48.733 -> U1:921MHz RX CH1 | U3:922MHz TX CH3 | ACK+Failover
16:04:48.733 ->
16:04:48.733 -> [PREFS] Loaded parent=0x0003 alt=0x0000
16:04:48.733 -> [U1-921] Init... OK
16:04:48.816 -> [U3-922] Init... OK
16:04:48.862 ->
16:04:48.862 -> [CH_CONFIG] Discovery @ 922MHz...
16:04:48.862 -> [CH_CONFIG] Broadcast 1/5...
16:04:49.320 -> [CH_CONFIG] Listening 5 detik...
16:04:49.919 -> [CH_CONFIG] Resp 0x0003 RSSI=-112.0dBm
16:04:50.646 -> [CH_CONFIG] Resp 0x0003 RSSI=-112.0dBm
16:04:54.328 -> [CH_CONFIG] parent=0x0003(-112.0dBm) alt=0x0000
16:04:54.328 -> [PREFS] Saved parent=0x0003 alt=0x0000
16:04:54.631 -> [CH_HELLO] →0x0003
16:04:55.104 -> [CH_HELLO] Terkirim
16:04:55.104 -> [U1] Listening 921MHz CH1...
16:04:55.104 -> [U3] Listening 922MHz CH3...
16:04:55.104 ->
16:04:55.104 -> [CH2] Siap | parent=0x0003 alt=0x0000
16:04:55.104 ->
```

STEP 3

CH3/GW Merespon Config CH2

Event: [U3 CH_CONFIG] Req dari 0x0002

RSSI: -112.0dBm

Aksi: Respond ke 0x0002 (delay 400ms)

Resp ke-1: @ 16:04:49.871

Resp ke-2: @ 16:04:50.645

HB: rx=1 ok=0 WiFi=OK MQTT=OK

CH_HELLO: [U3] CH=0x0002 RSSI:-112.0dBm

16:04:49.319 -> [U3 CH_CONFIG] Req dari 0x0002 RSSI:-112.0dBm

16:04:49.319 -> [CH_CONFIG] Respond ke 0x0002 (delay 400ms)

16:04:49.871 -> [CH_CONFIG] Resp ke-1

16:04:50.645 -> [CH_CONFIG] Resp ke-2

16:04:54.929 -> [HB|120s] rx=1 ok=0 WiFi=OK MQTT=OK

16:04:55.103 -> [U3 CH_HELLO] CH=0x0002 RSSI:-112.0dBm

Alur Komunikasi:

CH2 → [CH_CONFIG Broadcast] → CH3/GW

CH3/GW → [Resp ke-1, ke-2] → CH2

STEP 4

CH1 Aktif & Melakukan Config

Address:	CH1 = 0x0001
Mode:	STAR:920MHz RX MESH:921MHz TX
Config:	Broadcast 1/5 @ 921MHz
Resp ke-1:	0x0002 RSSI=-95.0dBm
Resp ke-2:	0x0002 RSSI=-94.0dBm (TERBAIK ✓)
Parent:	0x0002 @ -94.0dBm
CH_HELLO:	→ 0x0002 → Terkirim ✓
Status:	CH1 Siap parent=0x0002

```
16:08:00.682 -> === CH1 (0x0001) ===
16:08:00.682 -> STAR:920MHz RX | MESH:921MHz TX | ACK+Failover+PREFS
16:08:00.719 ->
16:08:00.719 -> [PREFS] Loaded parent=0x0002 alt=0x0000
16:08:00.719 -> [U1-920] Init... OK
16:08:00.802 -> [U3-921] Init... OK
16:08:00.836 ->
16:08:00.836 -> [CH_CONFIG] Mencari parent terbaik @ 921MHz...
16:08:00.836 -> [CH_CONFIG] Broadcast 1/5...
16:08:01.305 -> [CH_CONFIG] Listening response 5 detik...
16:08:01.876 -> [CH_CONFIG] Response 0x0002 RSSI=-95.0dBm
16:08:02.647 -> [CH_CONFIG] Response 0x0002 RSSI=-94.0dBm
16:08:06.308 -> [CH_CONFIG] SELESAI: parent=0x0002 (-94.0dBm) alt=0x0000
16:08:06.308 -> [PREFS] Saved parent=0x0002 alt=0x0000
16:08:06.606 -> [CH_HELLO] → 0x0002
16:08:07.091 -> [CH_HELLO] Terkirim
16:08:07.091 -> [U1] Listening 920MHz...
16:08:07.091 -> [U3] Listening 921MHz...
16:08:07.091 ->
16:08:07.091 -> [CH1] Siap | parent=0x0002 alt=0x0000
16:08:07.091 ->
```

STEP 5

CH2 Merespon Config dari CH1

Event: CH_CONFIG Req dari CH=0x0001

RSSI: -100.0dBm

Aksi: Respond ke 0x0001 (delay 400ms)

Resp ke-1: @ 16:08:01.876

Resp ke-2: @ 16:08:02.647

CH_HELLO: dari CH=0x0001 RSSI:-100.0dBm

16:08:01.306 ->

16:08:01.306 -> [CH_CONFIG] Req dari CH=0x0001 RSSI:-100.0dBm

16:08:01.306 -> [CH_CONFIG] Respond ke 0x0001 (delay 400ms)

16:08:01.876 -> [CH_CONFIG] Resp ke-1

16:08:02.647 -> [CH_CONFIG] Resp ke-2

16:08:07.091 -> [CH_HELLO] dari CH=0x0001 RSSI:-100.0dBm

CH2 mencatat CH1 sebagai child node.

CH1 parent=0x0002 ✓ | HELLO Terkirim

CH1 → [Config Req @ 921MHz] → CH2 → [Resp ke-1, ke-2] → CH1 → [CH_HELLO] → CH2

STEP 6

CH3/GW Merespon Config & Hello CH1

Event: [U1 CH_CONFIG] Req dari 0x0001

RSSI: -109.0dBm

Aksi: Respond ke 0x0001 (delay 400ms)

Resp ke-1: @ 16:08:01.876

Resp ke-2: @ 16:08:02.647

CH_HELLO: CH=0x0001 RSSI:-109.0dBm

Dicatat: cluster=0x0001 parent=0x0002

HB: rx=5 ok=0 WiFi=OK MQTT=OK

```
16:08:01.306 ->  
16:08:01.306 -> [U1 CH_CONFIG] Req dari 0x0001 RSSI:-109.0dBm  
16:08:01.306 -> [CH_CONFIG] Respond ke 0x0001 (delay 400ms)  
16:08:01.876 -> [CH_CONFIG] Resp ke-1  
16:08:02.647 -> [CH_CONFIG] Resp ke-2  
16:08:07.091 -> [U1 CH_HELLO] CH=0x0001 RSSI:-109.0dBm  
16:08:07.091 -> → cluster=0x0001 parent=0x0002 dicatat  
16:08:24.928 -> [HB|330s] rx=5 ok=0 WiFi=OK MQTT=OK  
16:08:54.906 -> [HB|360s] rx=5 ok=0 WiFi=OK MQTT=OK
```

GW mencatat: CH1 terhubung via CH2
Topologi: CH1 → CH2 → CH3/GW

Topologi Normal – Data Flow Berhasil



```
16:10:22.146 ->
16:10:22.146 -> +-----+
16:10:22.146 -> | CH_DATA_UP #21 dari CH=0x0002 (radio U3)
16:10:22.180 -> | RSSI:-103.0dBm SNR:8.0dB seq:5
16:10:22.180 -> +-----+
16:10:22.180 -> | Node:0xAA01 Temp:24.0C Hum:42.0% Batt:3247mV Light:478
16:10:22.180 -> | Sukses: 5/21
16:10:22.180 -> +-----+
16:10:22.180 -> [MQTT] Publish OK
16:10:22.180 -> [JSON] {"gatewayId":3,"radioSrc":3,"chId":2,"nodeId":43521,"temp":24,"hum":42,"battMv":3247,"light":478,"rssi":-103,"snr":8,"seq":5,"rxOk":1}
16:10:24.901 -> [HB/450s] rx=21 ok=5 WiFi=OK MQTT=OK
16:10:27.164 ->
16:10:27.164 -> +-----+
16:10:27.164 -> | CH_DATA_UP #24 dari CH=0x0002 (radio U3)
16:10:27.164 -> | RSSI:-103.0dBm SNR:8.2dB seq:6
16:10:27.164 -> +-----+
16:10:27.164 -> | Node:0xAA01 Temp:32.0C Hum:72.0% Batt:3143mV Light:166
16:10:27.164 -> | Sukses: 6/24
16:10:27.164 -> +-----+
16:10:27.208 -> [MQTT] Publish OK
16:10:27.208 -> [JSON] {"gatewayId":3,"radioSrc":3,"chId":2,"nodeId":43521,"temp":32,"hum":72,"battMv":3143,"light":166,"rssi":-103,"snr":8.25,"seq":6,"rxOk":1}
```

Skenario 2

Failover

CH2 Mati → CH1 Mencari Parent Baru



STEP 7

CH2 OFF → ACK Timeout (3x) → Failover!

Data Masuk dari Node

0xAA01 RSSI:-113dBm, Temp, Hum, Batt...

TX → 0x0002 attempt 1/3

ACK Timeout, backoff 100ms

TX → 0x0002 attempt 2/3

ACK Timeout, backoff 200ms

TX → 0x0002 attempt 3/3

ACK Timeout, backoff 400ms

[FAIL] failCnt=3

→ [FAILOVER] CH_CONFIG broadcast!

CH_CONFIG ulang

Mencari parent terbaik @ 921MHz...

```
16:11:30.369 ->
16:11:30.369 -> [STAR RX] node=0xAA01 RSSI:-113.0dBm SNR:2.5dB
16:11:30.369 -> Temp:35.0C Hum:77.0% Batt:3374mV Light:1014
16:11:30.507 -> [MESH TX] node=0xAA01 seq=17 -> 0x0002
16:11:30.507 -> [TX-0x0002 attempt 1/3]
16:11:32.862 -> [ACK] Timeout attempt 1, backoff 100ms
16:11:32.955 -> [TX-0x0002 attempt 2/3]
16:11:35.330 -> [ACK] Timeout attempt 2, backoff 200ms
16:11:35.517 -> [TX-0x0002 attempt 3/3]
16:11:37.888 -> [ACK] Timeout attempt 3, backoff 400ms
16:11:38.279 -> [FAIL] parent=0x0002 failCnt=3
16:11:38.279 -> [FAILOVER] CH_CONFIG broadcast!
16:11:39.267 ->
16:11:39.267 -> [CH_CONFIG] Mencari parent terbaik @ 921MHz...
16:11:39.267 -> [CH_CONFIG] Broadcast 1/5...
16:11:39.715 -> [CH_CONFIG] Listening response 5 detik...
16:11:40.314 -> [CH_CONFIG] Response 0x0003 RSSI:-105.0dBm
16:11:41.080 -> [CH_CONFIG] Response 0x0003 RSSI:-105.0dBm
16:11:44.751 -> [CH_CONFIG] SELESAI: parent=0x0003(-105.0dBm) alt=0x0000
16:11:44.751 -> [CH_CONFIG] Parent berubah: 0x0002 -> 0x0003
16:11:44.751 -> [PREFS] Saved parent=0x0003 alt=0x0000
16:11:45.017 -> [CH_HELLO] -> 0x0003
16:11:45.513 -> [CH_HELLO] Terkirim
16:11:45.513 -> [FAILOVER] Retry kirim dengan parent baru...
16:11:45.513 -> [TX-0x0003 attempt 1/3]
16:11:47.882 -> [ACK] Timeout attempt 1, backoff 100ms
16:11:48.012 -> [TX-0x0003 attempt 2/3]
16:11:48.617 -> [ACK] OK seq=17
```

STEP 8

CH3/GW Merespon Re-Config CH1

Event: [U1 CH_CONFIG] Req dari 0x0001

RSSI: -108.0dBm

Aksi: Respond ke 0x0001 (delay 400ms)

Resp ke-1: @ 16:11:40.329

Resp ke-2: @ 16:11:41.080

CH_HELLO: CH=0x0001 RSSI:-108.0dBm

Dicatat: cluster=0x0001 parent=0x0003 ✓

16:11:39.716 -> [U1 CH_CONFIG] Req dari 0x0001 RSSI:-108.0dBm

16:11:39.760 -> [CH_CONFIG] Respond ke 0x0001 (delay 400ms)

16:11:40.329 -> [CH_CONFIG] Resp ke-1

16:11:41.080 -> [CH_CONFIG] Resp ke-2

16:11:45.513 -> [U1 CH_HELLO] CH=0x0001 RSSI:-108.0dBm

16:11:45.513 -> → cluster=0x0001 parent=0x0003 dicatat

16:11:48.617 ->

16:11:48.617 -> |-----|

CH3/GW kini menjadi parent CH1

menggantikan CH2 yang sedang offline.

Parent berubah: 0x0002 → 0x0003

Parent CH1 Berubah: CH2 → CH3/GW

Parent Lama: CH2 (0x0002)

OFFLINE
ACK Timeout 3x → failCnt=3

↓ **FAILOVER**

✓ Parent Baru: CH3/GW (0x0003)

ONLINE
RSSI: -105.0dBm | ACK OK! seq=17

[CH1 Log - Failover]

```
[FAIL] parent=0x0002 failCnt=3  
[FAILOVER] CH_CONFIG broadcast!  
Resp 0x0003 RSSI=-105.0dBm (x2)  
[SELESAI] parent=0x0003(-105dBm)  
Parent berubah: 0x0002 → 0x0003  
[CH_HELLO] → 0x0003 → Terkirim  
[ACK] OK seq=17 ✓
```

Perubahan Topologi:

SEBELUM: Node → CH1 → CH2 → GW

SESUDAH: Node → CH1 —————→ GW

(melewati CH2, langsung ke CH3/GW)

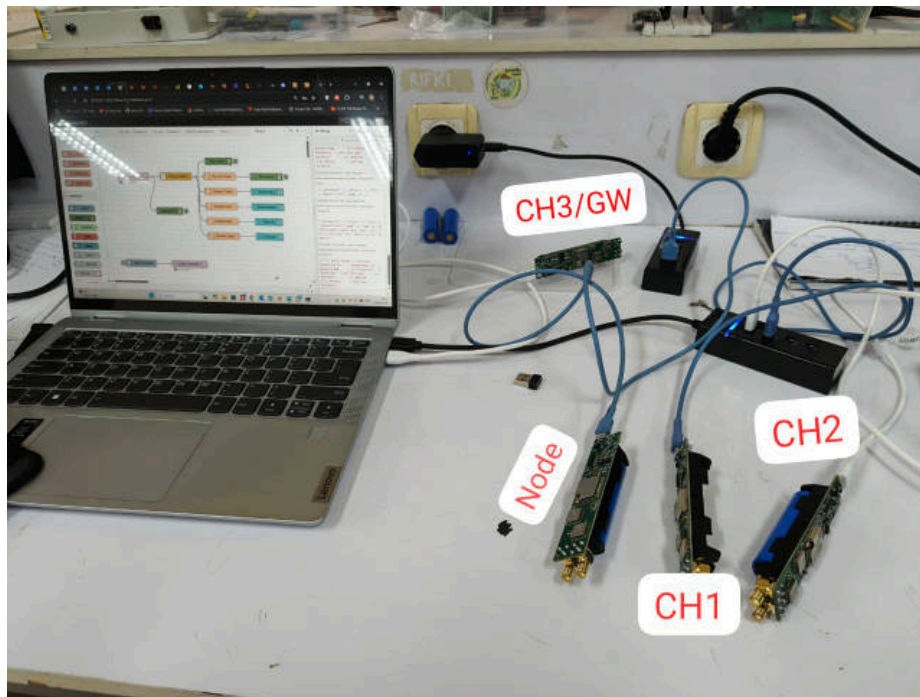
Data terkirim, tidak ada loss! ✓

Topologi Baru Setelah Failover – Data Flow



```
6:15:03.382 -> [U3_CH_CONFIG] Req dari 0x0002 RSSI:-110.0dBm
6:15:03.382 -> [CH_CONFIG] Respond ke 0x0002 (delay 400ms)
6:15:03.968 -> [CH_CONFIG] Resp ke-1
6:15:04.733 -> [CH_CONFIG] Resp ke-2
6:15:08.045 ->
6:15:08.045 -> +-----+
6:15:08.045 -> | CH_DATA_UP #102 dari CH=0x0001 (radio UI)
6:15:08.045 -> | RSSI:-111.0dBm SNR:3.8dB seq:58
6:15:08.045 -> +-----+
6:15:08.045 -> | Node:0xAA01 Temp:28.0C Hum:64.0% Batt:3266mV Light:591
6:15:08.092 -> | Sukses: 56/102
6:15:08.092 -> +-----+
6:15:08.092 -> [MQTT] Publish OK
6:15:08.092 -> [JSON] {"gatewayId":3,"radioSrc":1,"chId":1,"nodeId":43521,"temp":28,"hum":64,"battMv":3266,"light":591,"rsi":-111,"snr":3.75,"seq":58,"rxC
6:15:09.180 -> [U3_CH_HELLO] CH=0x0002 RSSI:-111.0dBm
6:15:13.062 ->
6:15:13.062 -> +-----+
6:15:13.062 -> | CH_DATA_UP #104 dari CH=0x0001 (radio UI)
6:15:13.062 -> | RSSI:-110.0dBm SNR:3.8dB seq:59
6:15:13.062 -> +-----+
6:15:13.062 -> | Node:0xAA01 Temp:31.0C Hum:78.0% Batt:3415mV Light:349
6:15:13.062 -> | Sukses: 57/104
```

Setup Fisik Pengujian



Kesimpulan

Pengujian Cluster Head – Gas Leak Detector

Konfigurasi Otomatis

Semua CH (CH1, CH2, CH3/GW) berhasil melakukan discovery dan memilih parent berdasarkan RSSI terbaik secara otomatis.

Topologi Star-Mesh Berfungsi

Data berhasil dikirim Node → CH1 → CH2 → CH3/GW dengan ACK confirmation setiap hop.

Failover Otomatis Sukses

Saat CH2 offline, CH1 otomatis deteksi timeout 3x, broadcast ulang, dan pindah parent ke CH3/GW.

Re-routing Transparan

Setelah failover, data tetap terkirim via jalur baru Node → CH1 → CH3/GW tanpa intervensi manual.

Kerjaan Minggu ke-2

27 Apr – 3 Mei 2026

01 Design.md (Algoritma Sistem)

In Progres

02 Uji Node GLD baru

✓ Done

03 Design Easy EDA Chamber

In Progres

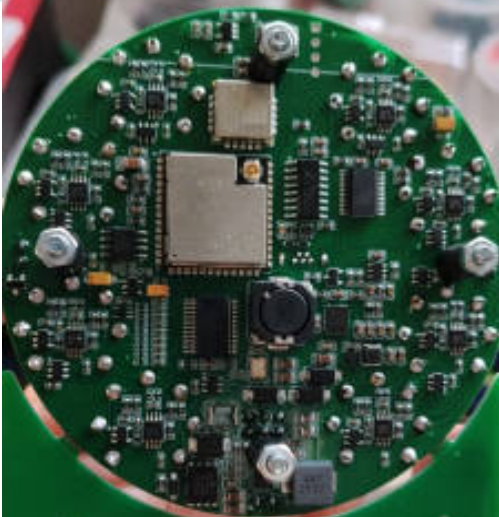
04 Uji Konsumsi baterai CH

✓ Done

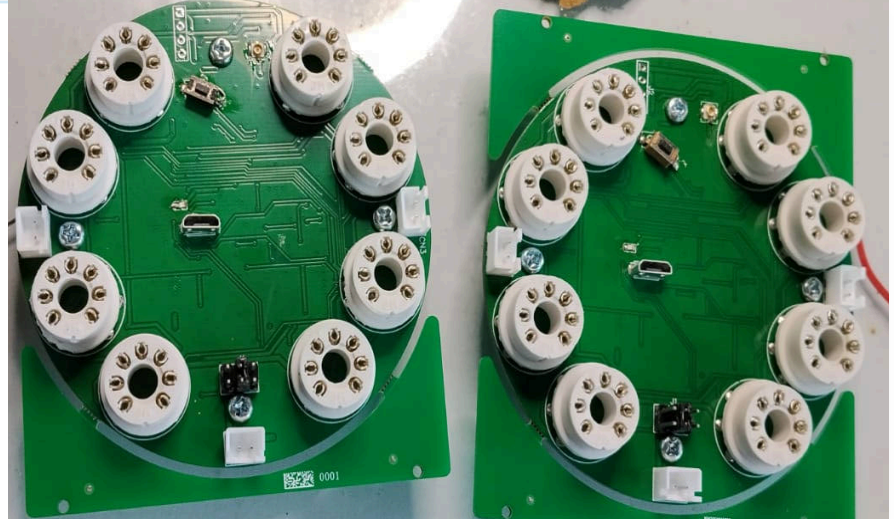
Uji Node GLD Baru

✓ Selesai

27 Apr – 3 Mei 2026



Board GLD Baru #0001 – Tampak Atas



2 Unit Node GLD Baru (#0001 & #0002) – Board Sensor



Hasil: Output Board GLD Baru semua parameter OK!

Desain hardware node baru telah berhasil diverifikasi. Semua output modul terkonfirmasi berjalan normal.

Uji Node GLD Baru – Output Serial

Verifikasi parameter ADC, LoRa, dan sensor pada board baru

✓ Selesai

27 Apr – 3 Mei 2026

```
[LOOP #910 | 00:15:15]
```

```
[ADC CH0] 1979706 | 0.009200 V | GAIN x64
[ADC CH1] 1866987 | 0.008676 V | GAIN x64
[ADC CH2] 1845150 | 0.008575 V | GAIN x64
[ADC CH3] 1549614 | 0.007202 V | GAIN x64
[ADC CH4] 1833502 | 0.008521 V | GAIN x64
[ADC CH5] 1728211 | 0.008032 V | GAIN x64
[ADC CH6] 1936471 | 0.008999 V | GAIN x64
[ADC CH7] 1845221 | 0.008575 V | GAIN x64
[LORA] -
[SYS] TCA=1 ADS=1 LORA=1 MCP=8/8
```

Terminal Output – Loop #910 | 00:15:15

Parameter Terverifikasi

TCA

TCA9548A I2C Multiplexer

1 ✓

ADS

ADS1115 ADC 16-bit

1 ✓

MCP

MCP3208 – 8 Channel ADC OK

8/8 ✓

ADC CH0–CH7 aktif

Tegangan CH0: 0.009V · Gain x64 · Semua channel terbaca normal

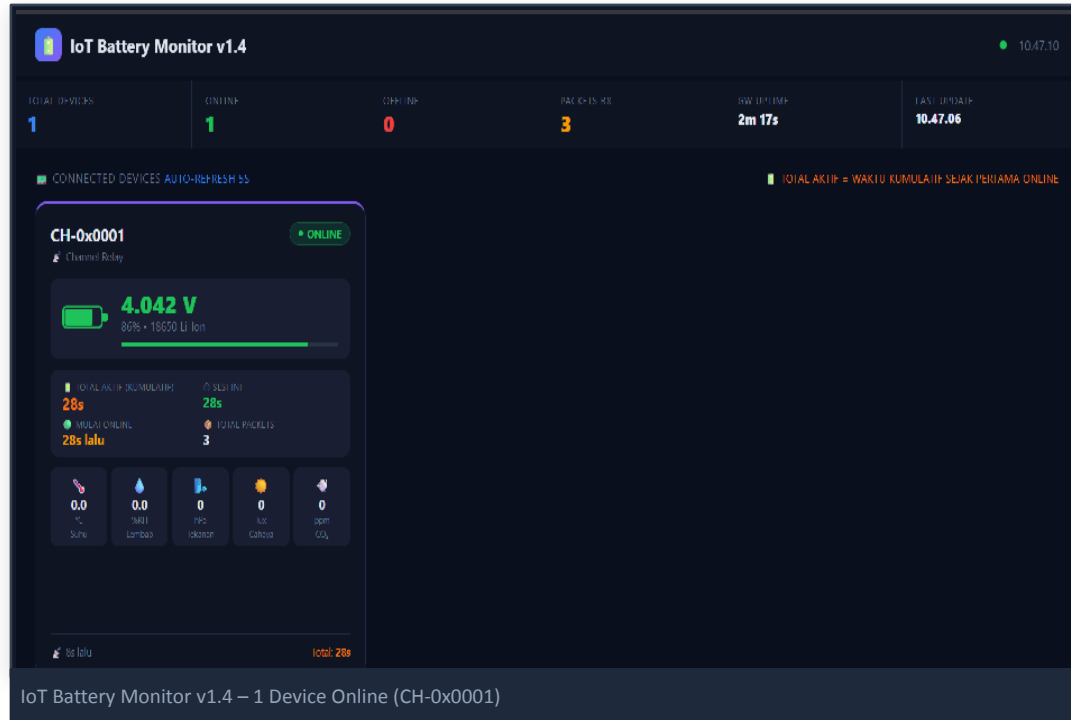
Sekarang sedang berlangsung proses warming up untuk memulai pembacaan sensor

Uji Konsumsi Baterai CH

✓ Selesai

27 Apr – 3 Mei 2026

Test 1: Satu device CH online – monitoring konsumsi baterai awal

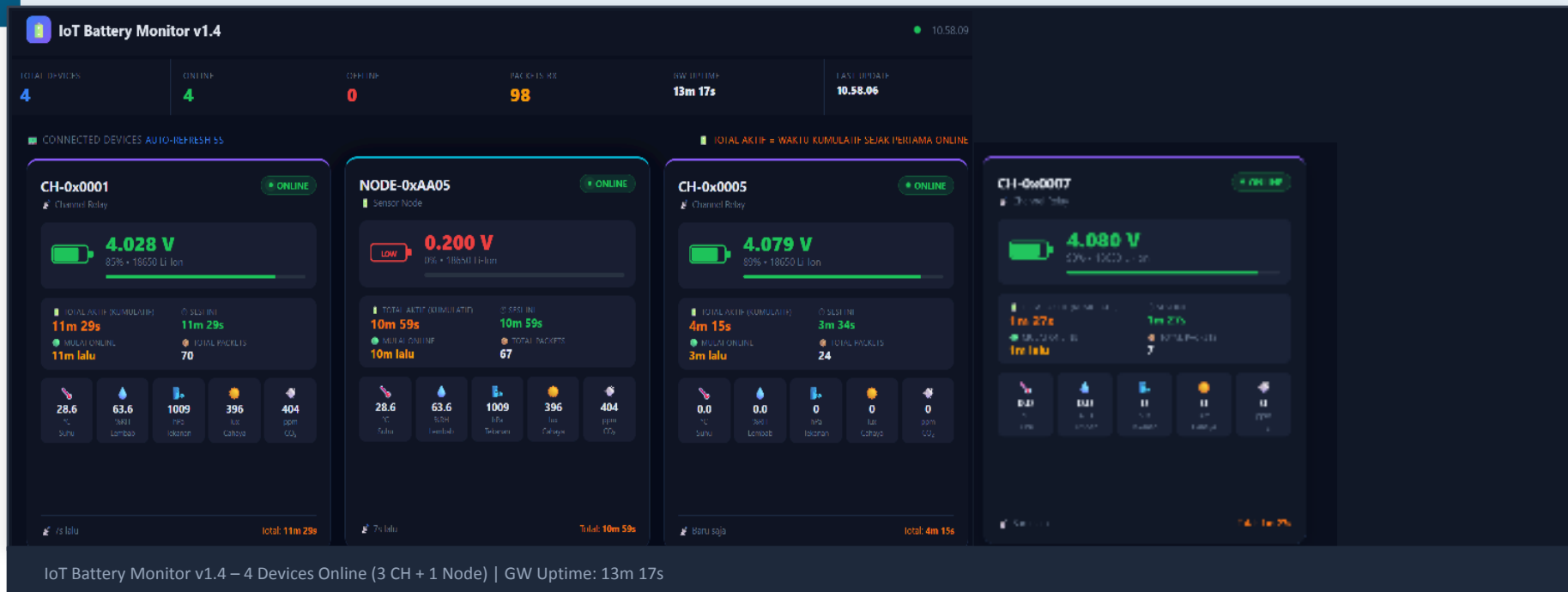


Uji Konsumsi Baterai CH

✓ Selesai

27 Apr – 3 Mei 2026

Test 2: 3 Device CH + 1 Node online – verifikasi multi-device



CH-0x0001

4.028 V · 85%

ONLINE

NODE-0xAA05

0.200 V · 0%

LOW BATT

CH-0x0005

4.079 V · 89%

ONLINE

CH-0x0007

4.080 V · 90%

ONLINE

Sekarang sedang sedang running untuk mengetahui berapa lama baterai habis

Kerjaan Minggu ke-3

01 Uji fungsional sistem (GLD-CH-GW-Server)

Pending

02 Pembuatan Chamber dan Uji Sensor Pembacaan Sensor

Pending

03 Uji Daya panel surya ke CH

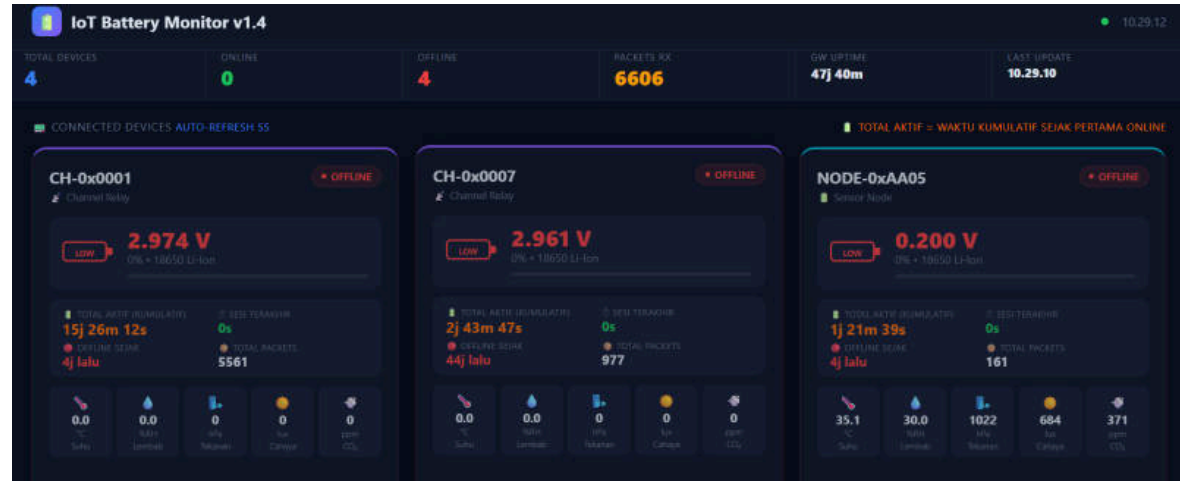
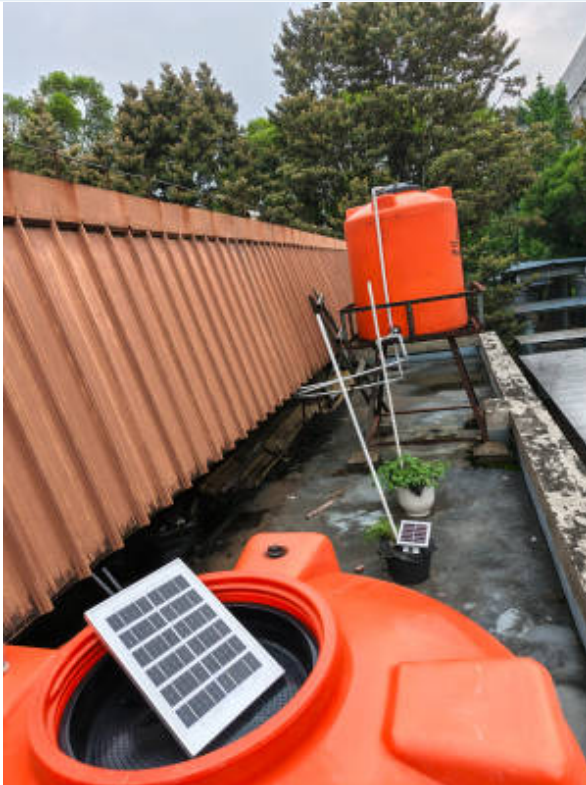
On Progress

Uji Konsumsi Panel Surya dan Baterai CH

On Progress

Test : 2 Device CH + 1 Node online – verifikasi multi-device

4 – 10 Mei 2026



Sekarang sedang sedang percobaan selama 1 Minggu

Uji Konsumsi Panel Surya dan Baterai CH

✓ Selesai

27 Apr – 3 Mei 2026

Test 2: 3 Device CH + 1 Node online – verifikasi multi-device



Sekarang sedang sedang percobaan selama 1 Minggu

Perbaiki firmware CH, menggunakan power recovery guard dan WDT



Kerjaan Minggu ke-4

01 Pengujian Sensor + ML

Pending

02 Uji jarak dan Reliabilitas komunikasi

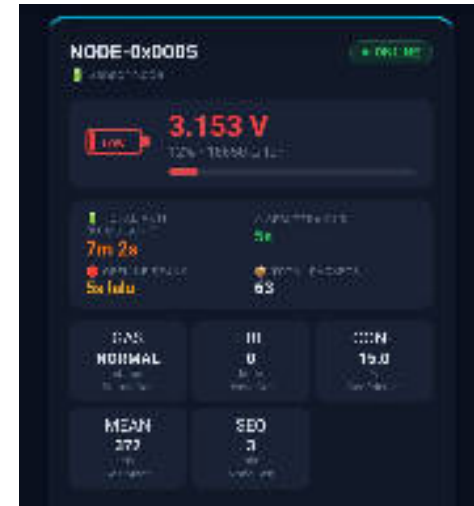
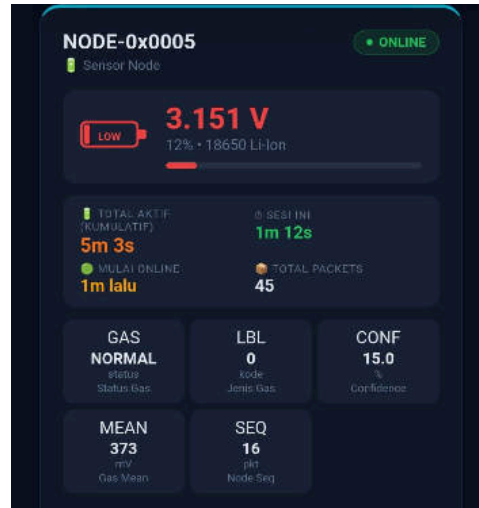
Pending

03 Uji Konsumsi baterai Node GLD

Pending



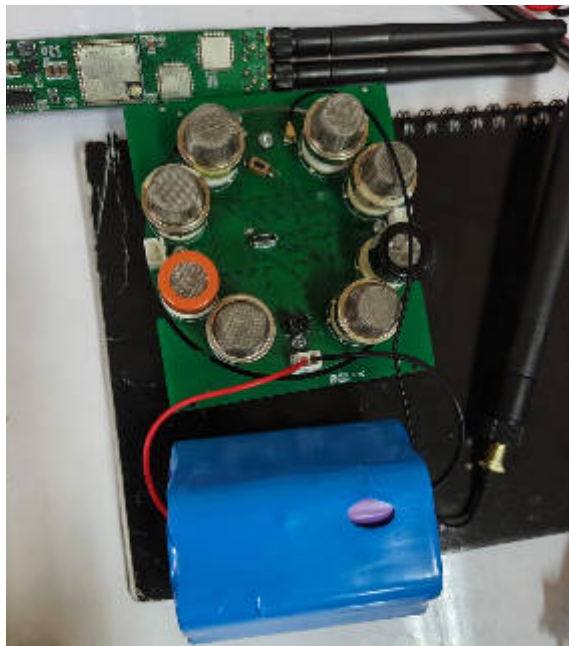
Uji Konsumsi baterai Node GLD



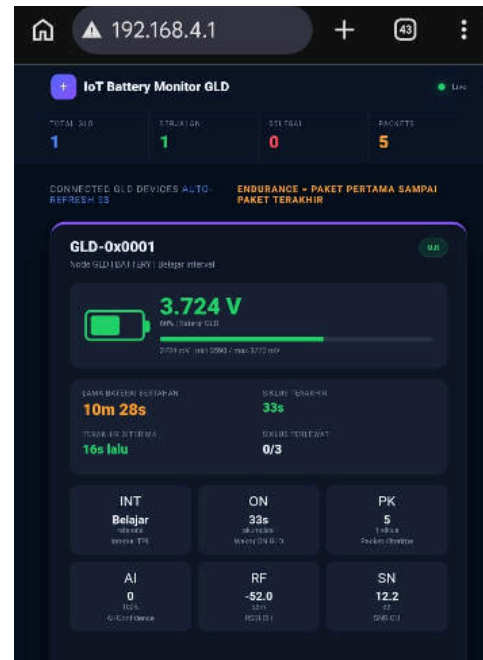
Node aktif selama 2 menit 37 detik, dan sleep selama 3 menit



Tegangan aktif 3.267V



Tegangan nonaktif 200mV



Waktu off 2 menit 37 detik

Uji Node GLD 24 V

Mode pembacaan sensor

```
15:20:57.797 > MQ8=0.00092, MQ135=0.00033, MQ3=0.00016, MQ5= 0.00128, MQ4=0.00026, MQ7= 0.00038, MQ6=0.00080, MQ2=0.00029
15:20:58.122 > MQ8=0.00092, MQ135=0.00033, MQ3=0.00016, MQ5=0.00128, MQ4=0.00026, MQ7=0.00038, MQ6=0.00080, MQ2=0.00029
15:20:58.448 > MQ8=0.00092, MQ135=0.00033, MQ3=0.00016, MQ5= 0.00128, MQ4=0.00026, MQ7= 0.00038, MQ6=0.00080, MQ2=0.00029
15:20:58.762 > MQ8=0.00092, MQ135=0.00033, MQ3=0.00016, MQ5=0.00128, MQ4=0.00026, MQ7=0.00038, MQ6=0.00080, MQ2=0.00029
15:20:59.088 > MQ8=0.00092, MQ135=0.00033, MQ3=0.00016, MQ5= 0.00128, MQ4=0.00026, MQ7= 0.00038, MQ6=0.00080, MQ2=0.00029
15:20:59.413 > MQ8=0.00092, MQ135=0.00033, MQ3=0.00016, MQ5=0.00128, MQ4=0.00026, MQ7=0.00038, MQ6=0.00080, MQ2=0.00029
15:20:59.689 > MQ8=0.00092, MQ135=0.00033, MQ3=0.00016, MQ5= 0.00128, MQ4=0.00026, MQ7= 0.00038, MQ6=0.00080, MQ2=0.00029
15:21:00.023 > MQ8=0.00092, MQ135=0.00033, MQ3=0.00016, MQ5=0.00128, MQ4=0.00026, MQ7=0.00038, MQ6=0.00080, MQ2=0.00029
15:21:00.332 > MQ8=0.00092, MQ135=0.00033, MQ3=0.00016, MQ5= 0.00128, MQ4=0.00026, MQ7= 0.00038, MQ6=0.00080, MQ2=0.00029
15:21:00.651 > MQ8=0.00092, MQ135=0.00033, MQ3=0.00016, MQ5=0.00128, MQ4=0.00026, MQ7=0.00038, MQ6=0.00080, MQ2=0.00029
15:21:01.019 > MQ8=0.00092, MQ135=0.00033, MQ3=0.00016, MQ5= 0.00128, MQ4=0.00026, MQ7= 0.00038, MQ6=0.00080, MQ2=0.00029
15:21:01.294 > MQ8=0.00092, MQ135=0.00033, MQ3=0.00016, MQ5=0.00128, MQ4=0.00026, MQ7=0.00038, MQ6=0.00080, MQ2=0.00029
15:21:01.660 > MQ8=0.00092, MQ135=0.00033, MQ3=0.00016, MQ5= 0.00128, MQ4=0.00026, MQ7= 0.00038, MQ6=0.00080, MQ2=0.00029
15:21:01.955 > MQ8=0.00092, MQ135=0.00033, MQ3=0.00016, MQ5=0.00128, MQ4=0.00026, MQ7=0.00038, MQ6=0.00080, MQ2=0.00029
15:21:02.301 > MQ8=0.00092, MQ135=0.00033, MQ3=0.00016, MQ5= 0.00128, MQ4=0.00026, MQ7= 0.00038, MQ6=0.00080, MQ2=0.00029
15:21:02.610 > MQ8=0.00092, MQ135=0.00033, MQ3=0.00016, MQ5=0.00128, MQ4=0.00026, MQ7=0.00038, MQ6=0.00080, MQ2=0.00029
```

Mode Running

```
15:22:11.664 -> [INTERFERENCE] class=0 conf=0.95 alarm=0 time=0ms
15:22:11.697 > [ATTACK] TX type=0x00 src=0x0001 dst=0x0004 seq=128 payload=14
15:22:13.075 -> [INTERFERENCE] RXF heuristic max=0.001276V sensor=3 class=0 conf=0.95
15:22:13.075 > [INTERFERENCE] class=0 conf=0.95 alarm=0 time=0ms
15:22:14.455 -> [INTERFERENCE] RXF heuristic max=0.001277V sensor=3 class=0 conf=0.95
15:22:14.455 > [INTERFERENCE] class=0 conf=0.95 alarm=0 time=0ms
15:22:15.833 -> [INTERFERENCE] RXF heuristic max=0.001279V sensor=3 class=0 conf=0.95
15:22:15.833 > [INTERFERENCE] class=0 conf=0.95 alarm=0 time=0ms
15:22:17.211 -> [INTERFERENCE] RXF heuristic max=0.001280V sensor=3 class=0 conf=0.95
15:22:17.222 > [INTERFERENCE] class=0 conf=0.95 alarm=0 time=0ms
15:22:18.589 -> [INTERFERENCE] RXF heuristic max=0.001280V sensor=3 class=0 conf=0.95
15:22:18.589 > [INTERFERENCE] class=0 conf=0.95 alarm=0 time=0ms
15:22:19.970 -> [INTERFERENCE] RXF heuristic max=0.001279V sensor=3 class=0 conf=0.95
15:22:19.970 > [INTERFERENCE] class=0 conf=0.95 alarm=0 time=0ms
15:22:21.355 -> [INTERFERENCE] RXF heuristic max=0.001278V sensor=3 class=0 conf=0.95
```



Output CH

```

15:11:49.887 > [HOUSEKEEP] batt-3578mV heap-362264 state-JOINED
15:12:12.764 -> [STAR_RX] node-0x0001 seq-116 alarm-0 ext-1 len-14
15:12:49.847 > [HOUSEKEEP] batt-3593mV heap-362264 state-JOINED
15:13:13.251 -> [STAR_RX] node-0x0001 seq-117 alarm-0 ext-1 len-14
15:13:49.850 > [HOUSEKEEP] batt-3581mV heap-362264 state-JOINED
15:14:13.697 -> [STAR_RX] node-0x0001 seq-118 alarm-0 ext-1 len-14
15:14:49.871 > [HOUSEKEEP] batt-3605mV heap-362264 state-JOINED
15:15:14.190 -> [STAR_RX] node-0x0001 seq-119 alarm-0 ext-1 len-14
15:15:49.871 > [HOUSEKEEP] batt-3554mV heap-362264 state-JOINED
15:16:14.660 -> [STAR_RX] node-0x0001 seq-120 alarm-0 ext-1 len-14
15:16:49.885 > [HOUSEKEEP] batt-3581mV heap-362264 state-JOINED
15:17:15.157 -> [STAR_RX] node-0x0001 seq-121 alarm-0 ext-1 len-14
15:17:49.890 > [HOUSEKEEP] batt-3563mV heap-362264 state-JOINED
15:18:15.626 -> [STAR_RX] node-0x0001 seq-122 alarm-0 ext-1 len-14
15:18:49.892 > [HOUSEKEEP] batt-3560mV heap-362264 state-JOINED
15:19:16.149 -> [STAR_RX] node-0x0001 seq-123 alarm-0 ext-1 len-14
15:19:49.861 > [HOUSEKEEP] batt-3563mV heap-362264 state-JOINED
15:20:16.659 -> [STAR_RX] node-0x0001 seq-124 alarm-0 ext-1 len-14
15:20:49.893 > [HOUSEKEEP] batt-3578mV heap-362264 state-JOINED
15:21:49.905 -> [HOUSEKEEP] batt-3575mV heap-362264 state-JOINED
15:22:11.697 > [STAR_RX] node-0x0001 seq-125 alarm-0 ext-1 len-14
15:22:49.866 -> [HOUSEKEEP] batt-3572mV heap-362264 state-JOINED
15:23:12.204 > [STAR_RX] node-0x0001 seq-126 alarm-0 ext-1 len-14
15:23:49.900 -> [HOUSEKEEP] batt-3578mV heap-362264 state-JOINED
15:24:12.678 > [STAR_RX] node-0x0001 seq-127 alarm-0 ext-1 len-14
15:24:49.872 -> [HOUSEKEEP] batt-3575mV heap-362264 state-JOINED
15:25:13.158 > [STAR_RX] node-0x0001 seq-128 alarm-0 ext-1 len-14
15:25:49.914 -> [HOUSEKEEP] batt-3572mV heap-362264 state-JOINED

```

Output GW

```

14:02:49.960 > [HELLO] ch-0x0064 parent-0x006f batt-3635mV
14:07:49.993 -> [HELLO] ch-0x0064 parent-0x006f batt-3590mV
14:12:49.999 > [HELLO] ch-0x0064 parent-0x006f batt-3548mV
14:17:50.022 -> [HELLO] ch-0x0064 parent-0x006f batt-3596mV
14:22:50.012 > [HELLO] ch-0x0064 parent-0x006f batt-3617mV
14:27:50.002 -> [HELLO] ch-0x0064 parent-0x006f batt-3575mV
14:32:50.026 > [HELLO] ch-0x0064 parent-0x006f batt-3605mV
14:37:50.022 -> [HELLO] ch-0x0064 parent-0x006f batt-3578mV
14:42:50.074 > [HELLO] ch-0x0064 parent-0x006f batt-3584mV
14:47:50.083 -> [HELLO] ch-0x0064 parent-0x006f batt-3590mV
14:52:50.070 > [HELLO] ch-0x0064 parent-0x006f batt-3590mV
14:57:50.087 -> [HELLO] ch-0x0064 parent-0x006f batt-3545mV
15:02:50.066 > [HELLO] ch-0x0064 parent-0x006f batt-3593mV
15:07:00.855 -> [6335/44] [E] [WebServer.cpp:630] _handleRequest(): request fa
15:07:50.107 > [HELLO] ch-0x0064 parent-0x006f batt-3593mV
15:12:50.127 -> [HELLO] ch-0x0064 parent-0x006f batt-3578mV
15:17:50.094 > [HELLO] ch-0x0064 parent-0x006f batt-3581mV
15:22:50.139 -> [HELLO] ch-0x0064 parent-0x006f batt-3575mV

```

Kerjaan Minggu ke-5

01 Uji Konsumsi baterai Node GLD

Pending

02 Uji sistem di lingkungan kampus

Pending

Bagian Teknis yang Perlu Dipastikan Saat Survei

Fokus: data engineering yang harus dibawa pulang agar DED, prototype, instalasi, dan field trial lebih terstruktur.

Pemetaan Area

Layout area • leak source candidate
Koordinat / referensi struktur
Kondisi lingkungan & akses maintenance

Lokasi Titik Node GLD

Point ID unik per kandidat node
Gas target, tinggi & orientasi sensor
Mounting, enclosure, clearance

Uji LoRa Awal / RF

RSSI, SNR, packet loss
Line-of-sight & obstruction
Kandidat CH, gateway

Power & Elektrik

24 VDC / battery / solar option
Jarak panel & jalur kabel
Grounding, backup gateway, risiko routing

Server & Jaringan

Lokasi gateway dan server lokal
LAN, IP, VLAN, firewall requirement
MQTT / API / dashboard route

Gas & Data AI

Target gas & existing detector
Threshold awal dan baseline normal
Test gas / dataset / cross-sensitivity

 **PERTIMBANGAN IMPLEMENTASI**

 **Coverage Area**
Pastikan area kritis tercover sesuai radius deteksi sensor

 **Cable Routing**
Rute kabel aman, rapi, dan terlindungi dari panas serta korosi

 **Power Availability**
Pastikan ketersediaan power stabil dan sesuai kebutuhan perangkat

 **Communication Reliability**
Pastikan sinyal stabil dan redundansi jaringan jika diperlukan

 **Maintenance Access**
Pastikan akses mudah dan aman untuk inspeksi dan perawatan

Metode kerja survei yang disarankan

Walkdown area bersama PIC RU → kodekan setiap kandidat titik → uji LoRa pada titik kritis → validasi power/server

Uji sistem di lingkungan kampus



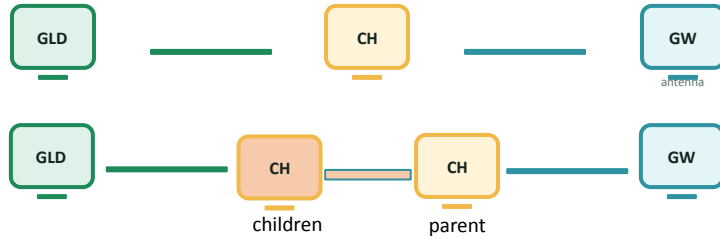
No	Kondisi / Lingkungan	Jarak (m)	Tinggi Ant. Tx (m)	Tinggi Ant. Rx (m)	RSSI (dBm)	PDR
1	area terbuka	10	3	3	-47	100%
2	area terbuka	30	3	3	-57	100%
3	area terbuka	100	3	3	-71	100%
4	Bawah pepohonan	80	3	3	-78	100%
5	bawah pepohonan	80	4	4	-84	100%
6	belakang gedung + pepohonan	-	3	3	-93	98%
7	belakang gedung + pepohonan (ant. diubah)	-	2	6	-81	100%
8	area Indomaret	-	2	6	-104	68%
9	area Fisika (batas jangkauan)	-	2	6	-112	26%

Pengaturan ketinggian antena berpengaruh terhadap kemampuan komunikasi data, namun perlu ditinjau kembali kondisi lingkungan, jika banyak vegetasi, mingkatkan tinggi antenna TX hingga mendekati dedaunan dan cabang pohon menyebabkan RSSI turun. Namun jika antenna TX diturunkan, dan antenna RX dinaikkan hingga 6 m, kemampuan komunikasi menjadi lebih baik (area Gedung dan pepohonan)

Pengujian simulasi komunikasi LoRa di sekitar ITB

Simulasi dilakukan sebelum site survey agar tim punya baseline jarak, kualitas link, failover, dan ketahanan CH outdoor.

Topologi uji: GLD -> CH -> GW



PDR

packet received / sent

RSSI

kekuatan sinyal

SNR

kualitas sinyal vs noise

Catatan: titik uji ITB dibuat mewakili skenario indoor, indoor dengan obstacle pohon/gedung, dan outdoor terbuka.

1. Uji paket data dan jarak

GLD dummy beberapa titik. Catat jarak GLD-CH-GW, RSSI, SNR, PDR, latency, dan obstacle.

2. Uji failover CH

Matikan beberapa CH. Pastikan A-CH mencari parent baru dan data tetap sampai ke GW.

3. Uji daya CH outdoor

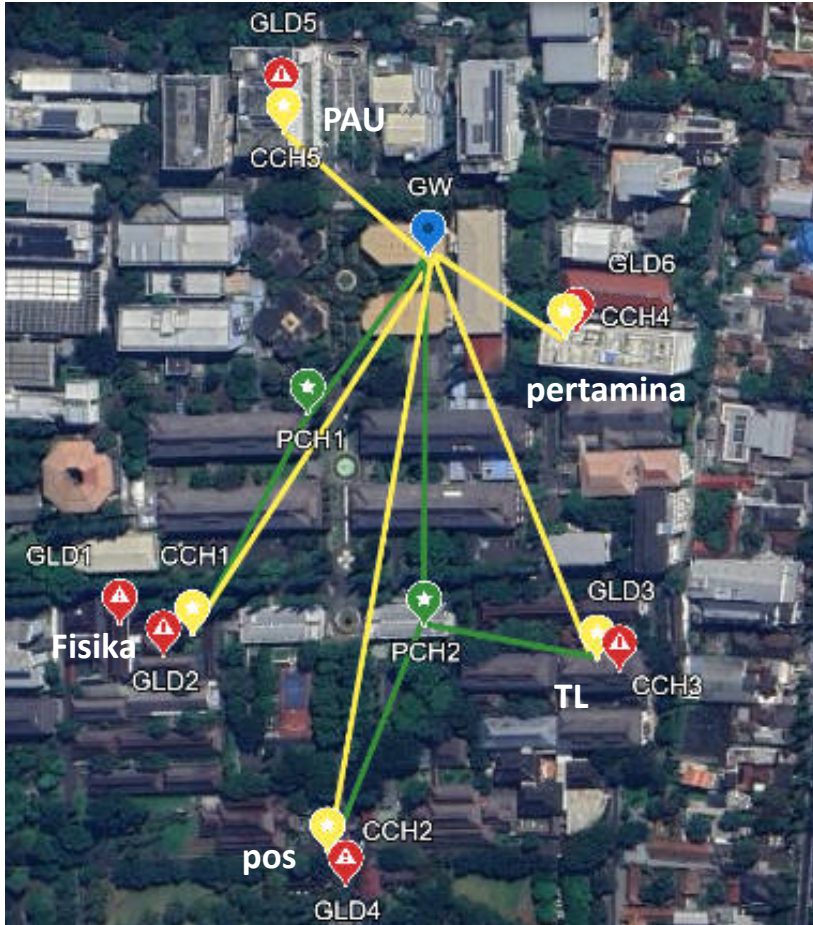
Nyalakan CH dengan solar/baterai, enclosure outdoor, dan logging minimal satu minggu.

Parameter yang dicatat per test point

- Koordinat TX/RX dan elevasi.
- LOS/NLOS, obstacle gedung/pohon/beton.
- Konfigurasi radio: SF, BW, TX power, antenna.
- Packet sent, received, PDR, RSSI, SNR, retry.

Mapping area menentukan titik GLD dan CH

Mapping dilakukan dari sumber risiko kebocoran ke desain komunikasi: titik GLD mengikuti sumber gas, titik CH mengikuti bukti kualitas RF.



GLD dalam ruangan area gedung

Prioritas di dekat sumber kebocoran: valve, flange, regulator, storage, pipa rawan vibrasi.

GLD dalam ruangan + banyak pohon

Perlu uji NLOS, multipath, dan posisi CH yang tidak tertutup kanopi/struktur.

GLD luar ruangan

Perhatikan enclosure, paparan hujan/panas, korosi, vandal risk, dan akses maintenance.

Logika penentuan CH

RSSI stabil

SNR aman

PDR 100%

CH final dipilih dari titik yang memberi margin RF terbaik, bukan hanya jarak terdekat. Jika PDR turun saat ada obstacle, kandidat CH harus digeser atau ditambah repeater/CH perantara.

Mekanisme pengujian

Tujuan uji: mencari jarak terjauh yang masih menjaga PDR 100% dengan RSSI/SNR yang punya margin aman, sambil mencatat obstacle dan koordinat.

1

GLD kirim paket ke CH

Kirim N paket per titik. Catat packet sent, packet received, PDR, RSSI, SNR, retry, dan latency.

2

Geser titik dan ulangi test

Pindahkan GLD bertahap, catat jarak dan obstacle: gedung, dinding, pohon, tangki, pipa, elevasi.

3

Tentukan batas link aman

Jarak terjauh dipilih hanya jika PDR 100% dan RSSI/SNR masih punya margin

4

Uji failover dan daya

Matikan CH tertentu untuk lihat parent baru; setelah itu nyalakan sistem outdoor terus menerus sampai 1 minggu.

cek RSSI/SNR/PDR saat CH di 5-6 m

CH mengaktifkan WiFi AP lokal + halaman HTML dashboard saat mode survei. Laptop/smartphone konek ke WiFi CH dari bawah, lalu membaca RSSI, SNR, PDR, packet counter, dan log test secara real-time.

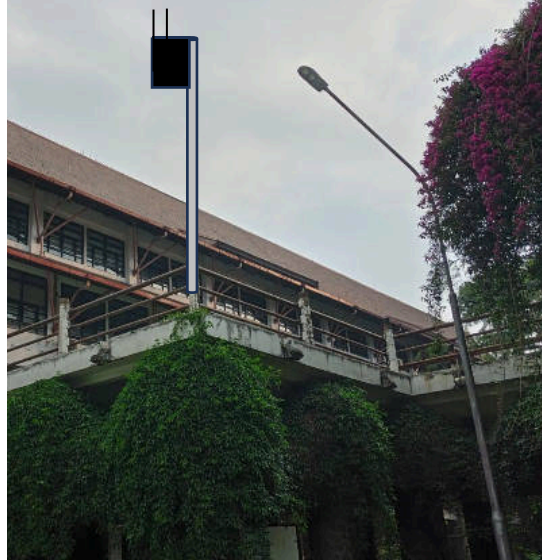
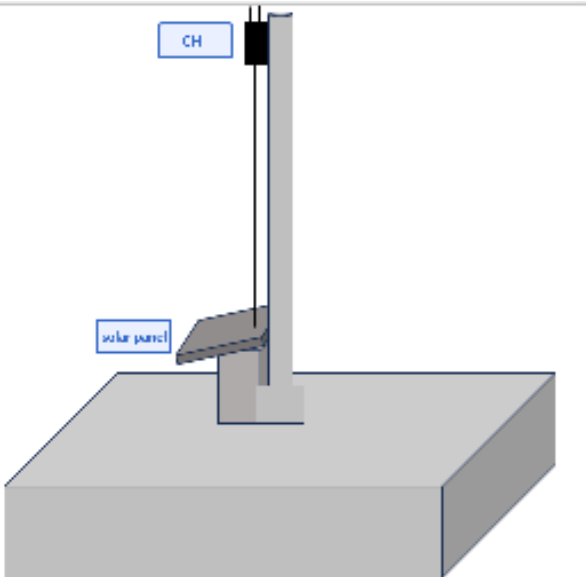
Tidak perlu USB panjang

Real-time dari bawah

Cocok untuk survei

Catatan deployment: WiFi AP debug hanya aktif pada mode survei/maintenance, memakai password, timeout otomatis, dan dimatikan saat operasi normal.

Peletakan node CH di rooftop



Rekomendasi dudukan

Diskusi dengan pak Totong untuk simulasi di ITB, di lapangan RU disesuaikan kembali

Stabilitas tiang 6 m

Periksa beban angin dan batas rooftop.

Penempatan panel surya

Panel kecil di bracket rendah pada base/ballast, bukan di puncak tiang, agar momen angin tidak bertambah besar.

Penempatan CH

CH IP65/IP66 di ujung tiang atau sedikit di bawah ujung tiang

Catatan HSE/struktur

Desain final harus dikonfirmasi oleh PIC fasilitas/struktur. Cek akses rooftop, proteksi petir/grounding, jalur kabel, dan izin kerja di ketinggian.

Output pengujian agar sistem layak dibawa ke site survey

Sebelum tim masuk RU, simulasi harus menghasilkan bukti teknis yang bisa dipakai sebagai dasar mapping, DED, dan eksekusi pemasangan.

RF

RF baseline per skenario area

Minimal indoor, indoor + obstacle, dan outdoor. Ada tabel jarak, RSSI, SNR, PDR, dan konfigurasi radio.

MAP

Topologi GLD-CH-GW awal

Ada candidate parent, rute failover, dan keputusan perlu CH tambahan/repeater.

FO

Bukti failover

Saat CH dimatikan, node/CH lain menemukan parent baru dan data sampai ke GW.

PWR

Bukti daya outdoor

CH menyala stabil dengan solar/baterai/enclosure minimal 1 minggu atau punya rencana uji lanjut.

DOC

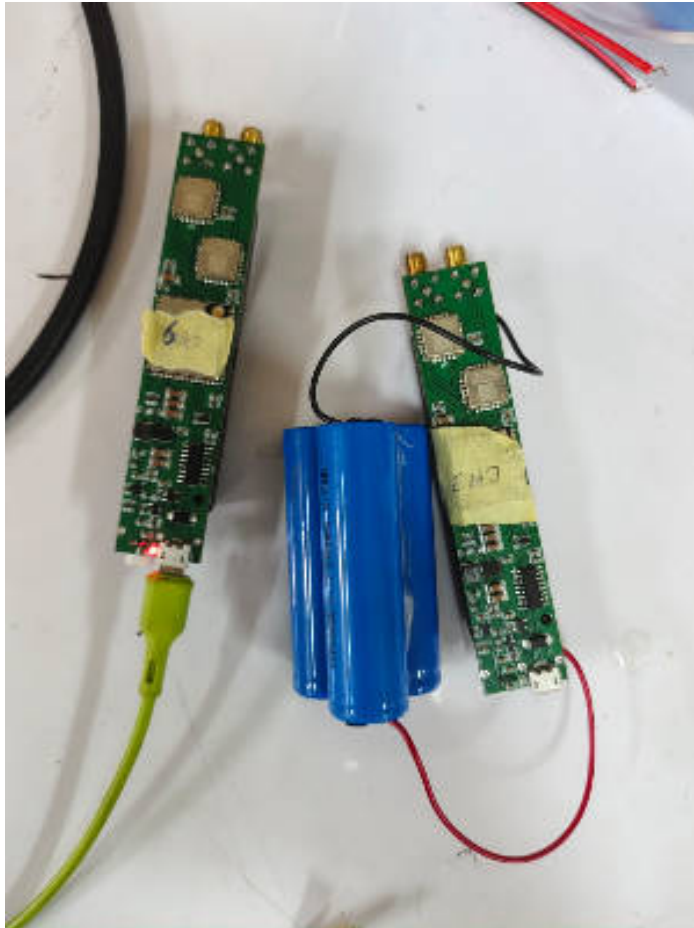
Dokumentasi lapangan siap

Template log, foto, koordinat, checklist, dan acceptance threshold siap dibawa ke RU.

Kriteria utama layak dibawa ke RU

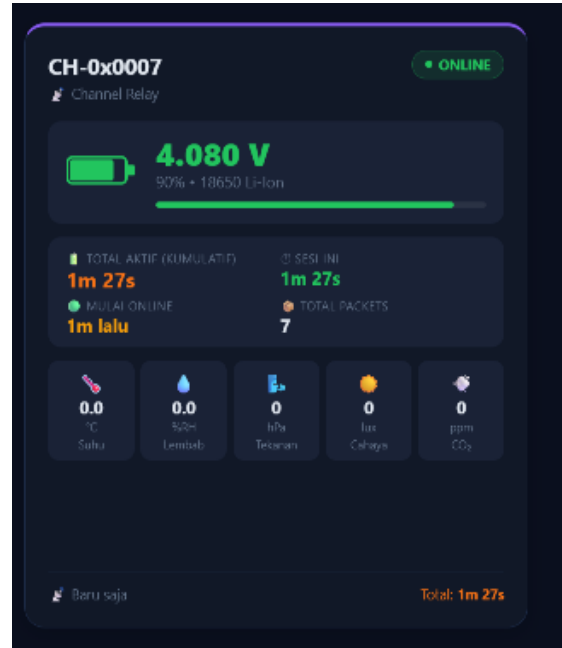
- PDR target 100% pada jarak kerja yang dipilih.
- RSSI/SNR tidak berada di batas minimum; harus punya margin.
- Failover terbukti, bukan hanya diasumsikan dari desain.
- Power CH outdoor punya rencana guard, logging, dan proteksi.

Uji 3 Baterai CH



3 baterai mampu supply board hingga 33 jam dengan periode transmit 10 detik

Uji 1 Baterai CH (LitoKala)

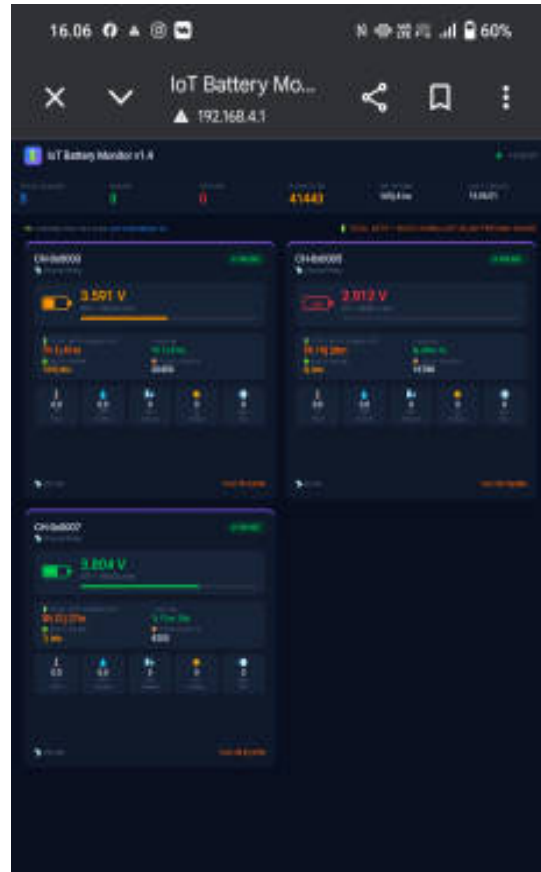


1 baterai LitoKala mampu supply board hingga 40 jam dengan periode transmit 10 detik

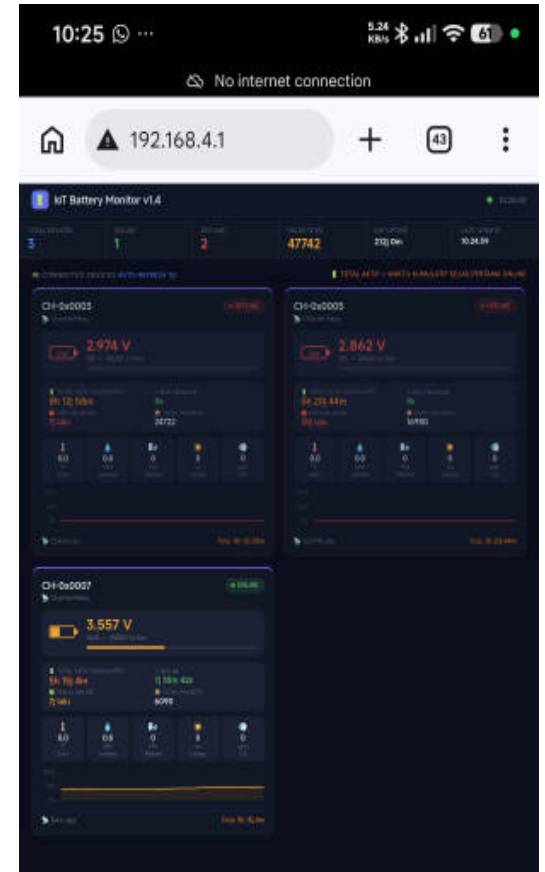
Uji 1 Baterai CH (LitoKala)



6 Juli
2026

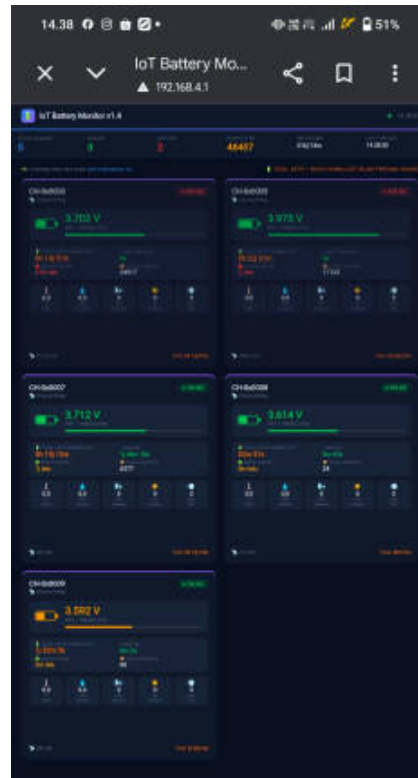


7 Juli
2026



8 Juli
2026

Uji 1 Baterai CH (LitoKala)



Analisis konsumsi daya CH

Observasi

CH5 hidup 5 hari; baterai awal 3.8 V

CH3 hidup 8 hari; baterai awal 4.0 V

Panel terukur Voc 7.77 V dan Isc 0.74 A

Makna teknis

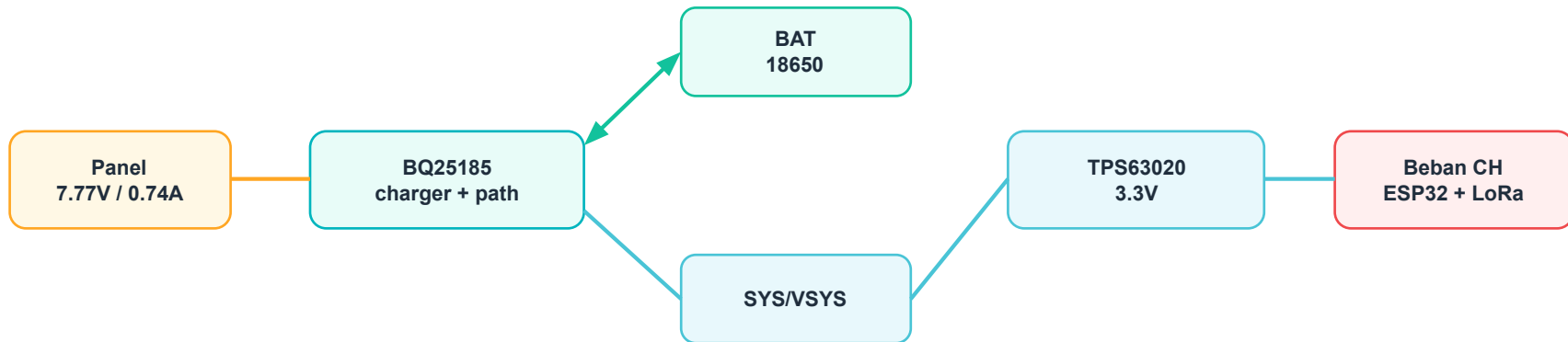
Panel memberi kontribusi energi yang hampir cukup

Rata-rata energi harian masih negatif

Baterai awal lebih penuh membuat runtime lebih panjang

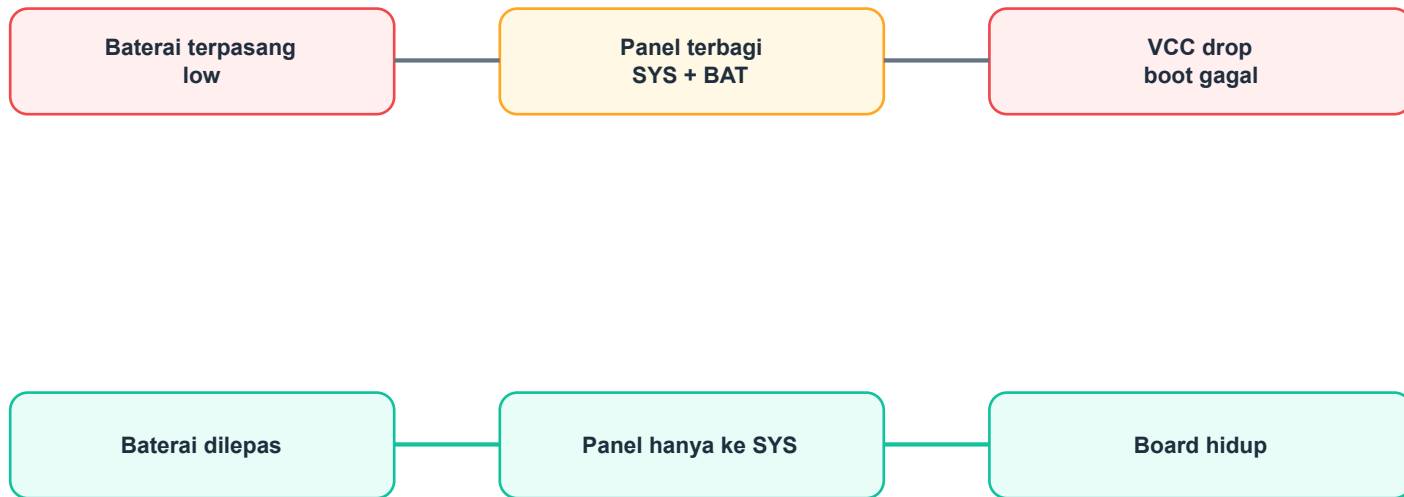
Kesimpulan awal: firmware sistem sudah betjalan dengan baik, tetapi terdapat defisit energi kecil + power-path tidak stabil saat baterai low

Alur penggunaan daya CH



Saat baterai low, arus panel terbagi antara mengisi baterai dan menyalakan SYS; jika tidak cukup, VCC bisa drop dan sistem bisa hang.

Alur penggunaan daya CH



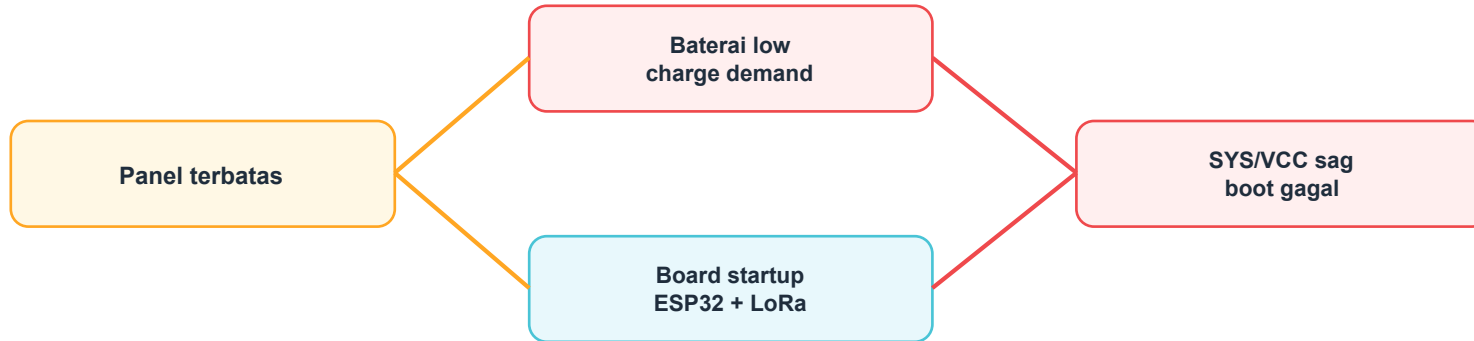
Alur penggunaan daya CH

Saat panel cukup

SYS memberi power ke board
BAT mengisi baterai dari sisa arus
Sistem tidak harus menunggu baterai penuh

Saat panel terbatas

Input current dibatasi
Baterai low menarik arus charge
SYS bisa turun jika total beban terlalu besar



Konsumsi Arus CH

Komponen	Mode	Arus dipakai
ESP32-S3	aktif firmware CH	58.0 mA
LoRa U1 E22	RX/listen 920 MHz	6.8 mA
LoRa U3 E22	RX/listen 921.7 MHz	6.8 mA
CH340	standby/powered	7.0 mA
LED	aktif ringan	4.0 mA
TX LoRa avg	17 dBm, tiap 30 detik	1.4 mA
Total rata-rata	listen + transmit periodik	84.0 mA

Konsumsi energi harian
Tegangan sistem 3.3V, arus rata-rata 84 mA. Maka:

 $P = 3.3V \times 0.084A = 0.2772\text{ W}$

 $E = P \times 24\text{ jam} = 0.2772W \times 24$
 $E = 6.65\text{ Wh/hari}$

Efisiensi ~85-90%

$E_{VSYS} = 6.65 / 0.88$
 $= 7.56\text{ Wh/hari}$

Device	V awal	V akhir	Energi turun	Defisit/hari
CH5	3.8 V	3.2 V	~3.4 Wh	~0.69 Wh/hari
CH3	4.0 V	3.3 V	~5.0 Wh	~0.62 Wh/hari

Energi panel surya

Pengukuran panel

$V_{oc} = 7.77 \text{ V}$

$I_{sc} = 0.74 \text{ A}$

$V_{oc} \times I_{sc} = 5.75 \text{ W}$

$P_{max} = V_{oc} \times I_{sc} \times \text{fill factor (65-80\%)} = 3.74 \text{ W}$ sampai 4.60 W

Estimasi efektif

P_{max} realistis: $3.7\text{-}4.6 \text{ W}$

Peak sun hour: 3-4 jam

$P_{max} \times 3\text{-}4 \text{ jam}$

Panel mentah
11-18 Wh/hari

(45-50%)

Pengaruh
charger + cuaca

Panel mentah x (45-50%)

Efektif
5-9 Wh

Beban CH
~ 8 Wh

Estimasi Energi Panel Surya

estimasi panel efektif

$E_{\text{panel}} = E_{\text{beban}} - \text{defisit} = 7.56 - 0.65 = 6.91 \text{ Wh/hari}$

Periode	Durasi	Energi panel	Energi CH	Efek baterai
Panel efektif	4 jam	6.91 Wh	1.58 Wh	+5.33 Wh
Panel lemah/malam	20 jam	0 Wh	5.99 Wh	-5.99 Wh
Total 24 jam	24 jam	6.91 Wh	7.56 Wh	-0.66 Wh

Estimasi pengisian baterai LitoKala 4000 mAh

Charge dibatasi ISET dan harus berbagi daya dengan beban CH

Parameter	Nilai
Kapasitas baterai	4000 mAh = 4 Ah
Energi nominal	$3.7 \text{ V} \times 4 \text{ Ah} = 14.8 \text{ Wh}$
ISET sekarang	1 kOhm
Fast charge BQ25185	~300 mA
Waktu ideal tanpa beban	$4000 / 300 = 13.3 \text{ jam}$
Dengan fase CV	~16-18 jam charge efektif

Jika panel efektif 5 jam/hari

Tanpa beban: 3-4 hari
Sambil CH aktif: sekitar 3-4 hari
Dari 3.2-3.3 V: sekitar 2-3 hari
Mendung bisa jadi 5-7 hari

Waktu charge bukan hanya kapasitas baterai; ditentukan oleh arus ISET, energi panel efektif, dan beban CH yang tetap hidup.

ISET lebih besar punya batas panel

Charge lebih cepat dapat membuat panel drop saat matahari tidak kuat

RISET	ICHG kira-kira	Catatan
1.0 kOhm	300 mA	konservatif saat ini
750 Ohm	400 mA	layak diuji
620 Ohm	480-500 mA	batas atas masuk akal
300 Ohm	1000 mA	tidak disarankan untuk panel ini

Risiko jika ICHG terlalu besar

Panel tertarik turun, VSYS drop, ESP32 reset/hang, dan LoRa gagal init.

Rekomendasi uji

Coba 400 mA dulu, ukur input panel dan VSYS dengan INA219 sebelum naik ke 500 mA.

Solusi Penyediaan Energi untuk CH

1. Tambah buffer

3x18650 paralel
Sag lebih kecil
Startup lebih kuat
Runtime lebih panjang

2. Tambah Energi Harian

Panel lebih besar
Atau 2 panel paralel
Orientasi dan waktu pencahayaan
efektif lebih baik

Target energi masuk > beban

Rekomendasi praktis: 3x18650 paralel + panel lebih besar/2 panel

Validasi lapangan perlu mengukur energi panel langsung

INA219 dipasang seri di jalur positif panel sebelum CN3 SOLAR

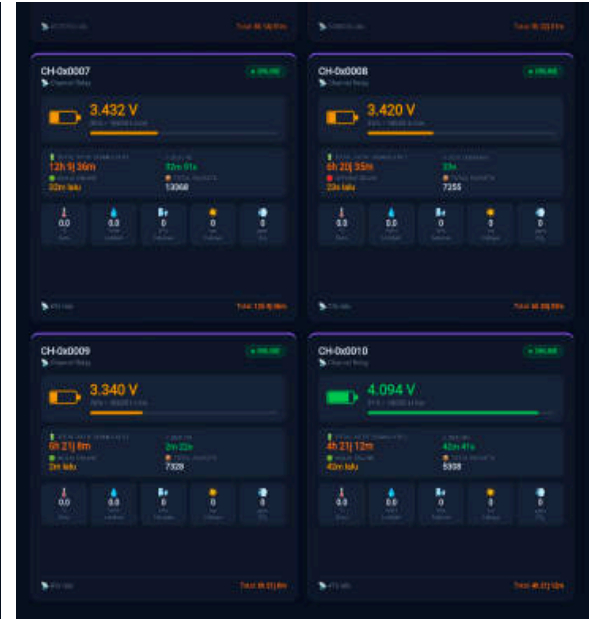
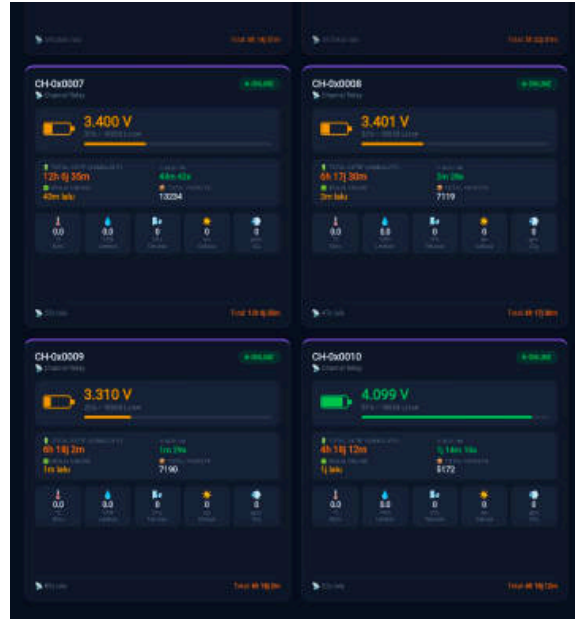
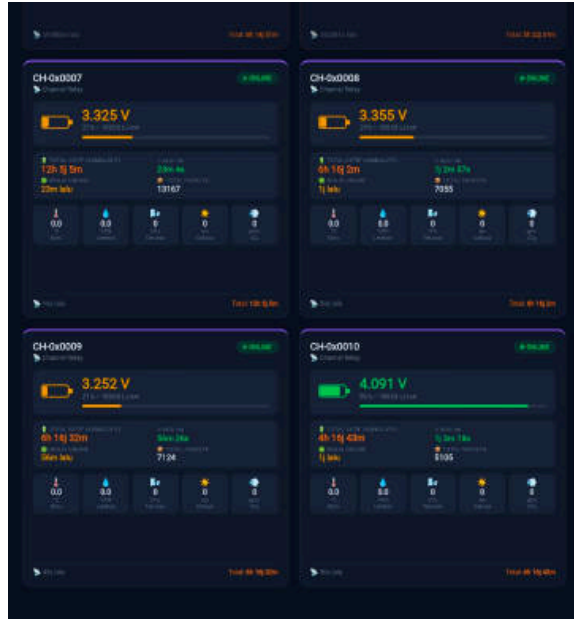


Yang dicatat

Log V_{panel} , I_{panel} , $P = V \times I$, lalu akumulasi $Wh += P \times \text{delta_waktu_jam}$ dari pagi sampai sore.

Performa CH dengan supply dari 2 panel surya

16 Juli 2026



Parameter	Nilai	Keterangan
Voc	7.77 V	tegangan open-circuit
Isc	1.29 A	arus short-circuit aktual
Voc x Isc	~10.02 W	daya mentah
Pmax (kerja)	~6.51 W	asumsi FF 65%

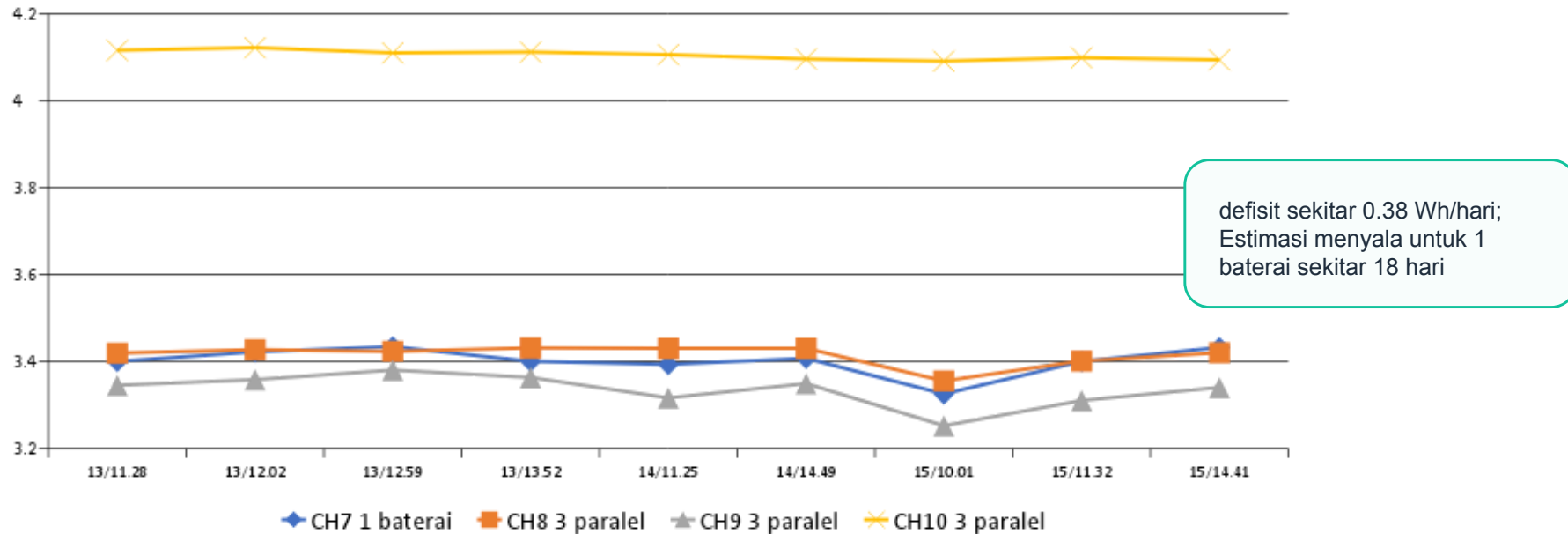
Panel efektif estimasi $\approx 9.1 - 12.0$ Wh/hari
Kebutuhan CH ≈ 7.56 Wh/hari
Surplus $\approx 1.5 - 4.4$ Wh/hari

Data tegangan baterai awal

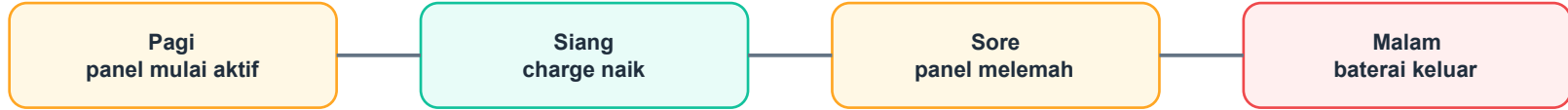
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Device	Konfigurasi baterai	Tegangan awal	Kondisi awal
CH7	1 baterai	3.77 V	menengah
CH8	3 paralel	3.67 V	lebih rendah
CH9	3 paralel	3.70 V	menengah
CH10	3 paralel	4.19 V	penuh

Grafik harian baterai CH



Analisis karakteristik power sistem CH



Tegangan baterai yang terlihat di dashboard adalah hasil gabungan state-of-charge, charging sesaat, beban CH, dan relaksasi baterai. Saat panel melemah, arus charge turun dan tegangan bisa kembali turun walaupun sebelumnya sempat naik.

Parameter	Nilai
ISET sekarang	1 kOhm
Fast charge kira-kira	300 mA
Energi ke 1 sel saat 3.4 V, 4 jam	$3.4V \times 0.3A \times 4h = 4.1 \text{ Wh}$
Konsumsi CH harian	~7.56 Wh/hari
3 baterai paralel	kapasitas 3x, waktu penuh juga lebih lama

Dua panel paralel membuat input cukup, tetapi arus BAT tetap dibatasi BQ25185.

RISET	ICHG	1 baterai 4000mAh	3 paralel 12000mAh	Catatan
1.0 kOhm	300 mA	16-18 jam efektif	48-54 jam efektif	sekarang
750 Ohm	400 mA	12-14 jam efektif	36-42 jam efektif	layak uji
600 Ohm	500 mA	10-11 jam efektif	30-33 jam efektif	batas aman awal
470 Ohm	640 mA	8-9 jam efektif	24-27 jam efektif	cek panas
300 Ohm	1000 mA	5-6 jam efektif	15-18 jam efektif	tidak disarankan langsung

ANALISIS DAYA SISTEM GLD

Simulasi Kebutuhan Baterai Node GLD

Variasi arus, waktu ON, kapasitas baterai, dan skenario target 30 hari

Perhitungan runtime GLD untuk 7 Baterai

Parameter	Nilai	Keterangan
Kapasitas	7 x 4000 mAh	28 Ah paralel
Energi nominal	28Ah x 3,7V	103,6 Wh
Usable	80%	margin proteksi baterai
Efisiensi	85%	rugi boost/regulator
Energi kerja	103,6 x 80% x 85%	70.4 Wh

Rumus utama

$\text{Runtime} = \text{Wh tersedia} / (5V \times I_{\text{ON}} \times \text{duty cycle}).$

$\text{Duty cycle} = T_{\text{ON}} / (T_{\text{ON}} + 159s).$

Input	Nilai
I_ON	1,15 A
T_ON	65 detik
T_OFF	159 detik
Baterai	7P / 28Ah
Model	Wh, usable 80%, efisiensi 85%

Daya ON

5.75 W

5V x 1,15A

Duty cycle

29.0%

65/(65+159)

Daya rata-rata

1.67 W

P_ON x duty

Runtime

1.76 hari

42.2 jam

Variasi waktu ON

I_ON 1,15 A, Baterai 7P, dan T_OFF 159 detik

ON	Duty	Pavg	Runtime
10s	5.9%	0.34 W	8.63 hari
20s	11.2%	0.64 W	4.57 hari
30s	15.9%	0.91 W	3.22 hari
40s	20.1%	1.16 W	2.54 hari
50s	23.9%	1.38 W	2.13 hari
60s	27.4%	1.58 W	1.86 hari

ON	Duty	Pavg	Runtime
70s	30.6%	1.76 W	1.67 hari
80s	33.5%	1.92 W	1.53 hari
90s	36.1%	2.08 W	1.41 hari
100s	38.6%	2.22 W	1.32 hari
110s	40.9%	2.35 W	1.25 hari
120s	43.0%	2.47 W	1.19 hari

OFF 159 detik, dan ON 10 detik masih hanya sekitar 8,63 hari. Target 30 hari perlu OFF jauh lebih panjang atau arus jauh lebih kecil.

Variasi jumlah baterai

I_ON 1,15 A, T_ON 65 detik, dan T_OFF 159 detik

Konfig.	Kapasitas	Energi nominal	Runtime
7P	28 Ah	103.6 Wh	1.76 hari
14P	56 Ah	207.2 Wh	3.52 hari
30P	120 Ah	444.0 Wh	7.54 hari
60P	240 Ah	888.0 Wh	15.08 hari
90P	360 Ah	1332.0 Wh	22.62 hari
120P	480 Ah	1776.0 Wh	30.16 hari

Dengan jadwal ON 65 detik dan arus 1.15A, membutuhkan 120 sel 4000 mAh paralel baru mendekati 30 hari.

Variasi Estimasi Arus ON

T_ON tetap 65 detik, T_OFF 159 detik, baterai 7P 28Ah

I_ON	P_ON @5V	Runtime	Runtime
0.64 A	3.20 W	75.9 jam	3.16 hari
0.80 A	4.00 W	60.7 jam	2.53 hari
1.00 A	5.00 W	48.6 jam	2.02 hari
1.15 A	5.75 W	42.2 jam	1.76 hari
1.50 A	7.50 W	32.4 jam	1.35 hari
1.70 A	8.50 W	28.6 jam	1.19 hari
2.00 A	10.00 W	24.3 jam	1.01 hari

Jika Target Minimal 30 hari (power baterai)

Strategi	Kebutuhan	Kondisi	Catatan
Tambah baterai	~120 sel 4000 mAh	477 Ah nominal	besar/berat
Turunkan arus ON	~67 mA	ON 65s, OFF 159s	tidak realistis untuk heater
Perpendek ON	~2.8 s	OFF tetap 159s	terlalu pendek untuk MQ
Perpanjang OFF	~62.6 menit	ON tetap 65s	opsi paling masuk akal

Kesimpulan

Untuk 7P dan 1,15 A, target 30 hari mensyaratkan interval OFF sekitar 63 menit bila waktu ON tetap 65 detik.

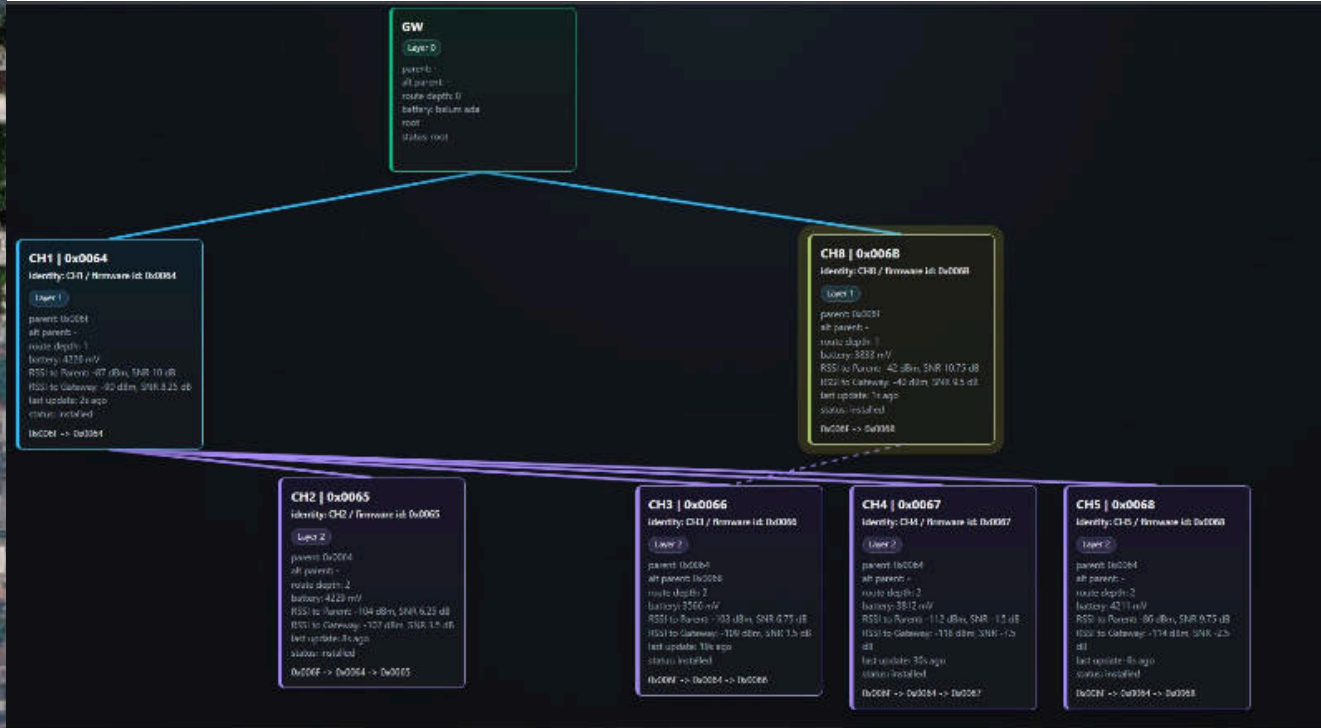
Catatan Tambahan

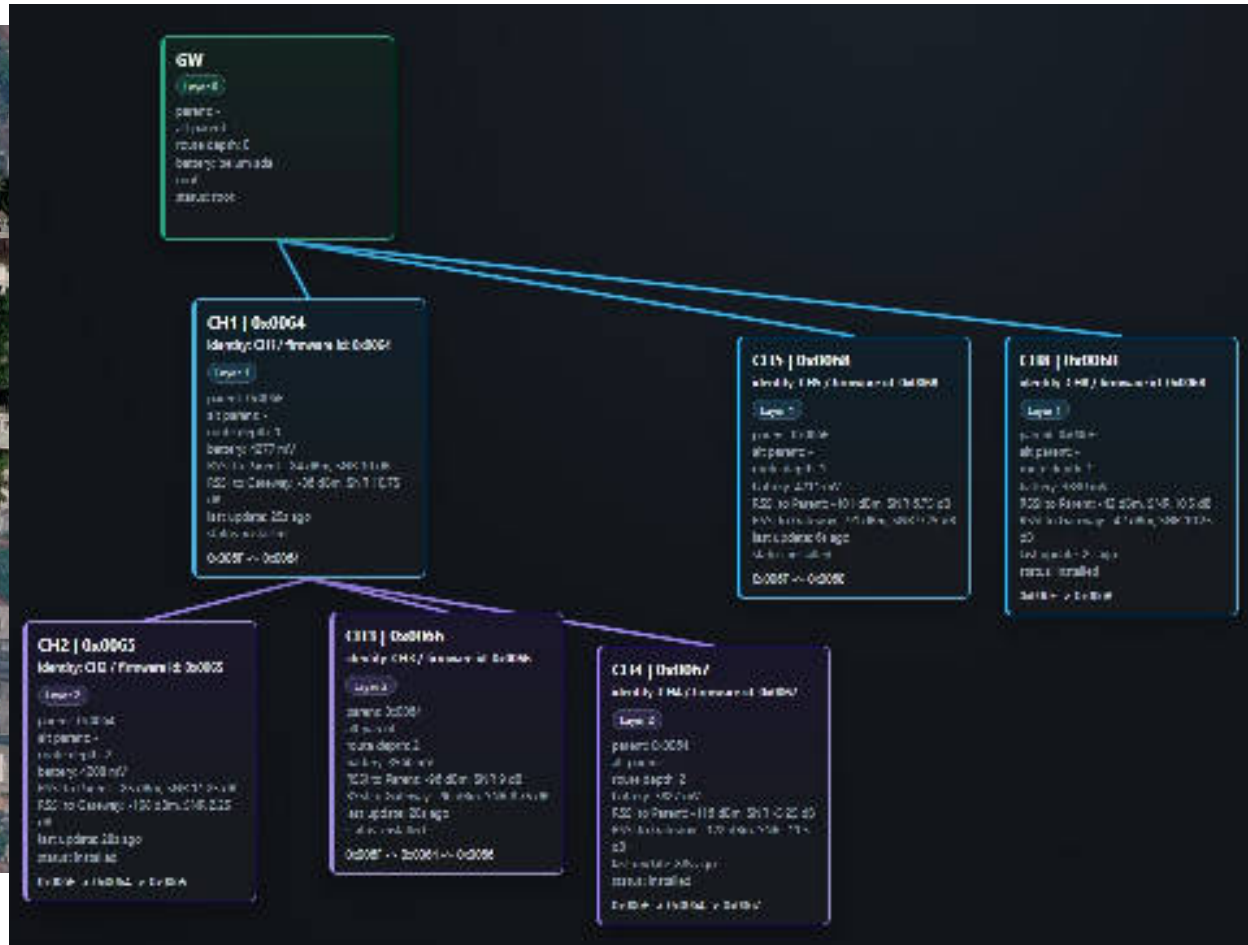
- Baterai LitoKala yang digunakan banyak yang hanya 2500 mAh, tidak sampai 4000 mAh
- TPL5010 masih belum berjalan optimal, saat sistem non-aktif, board masih mendapat tegangan ~0.4V dan masih aktif cuplik data

Uji Mesh Cluster Head

Uji Mesh 1

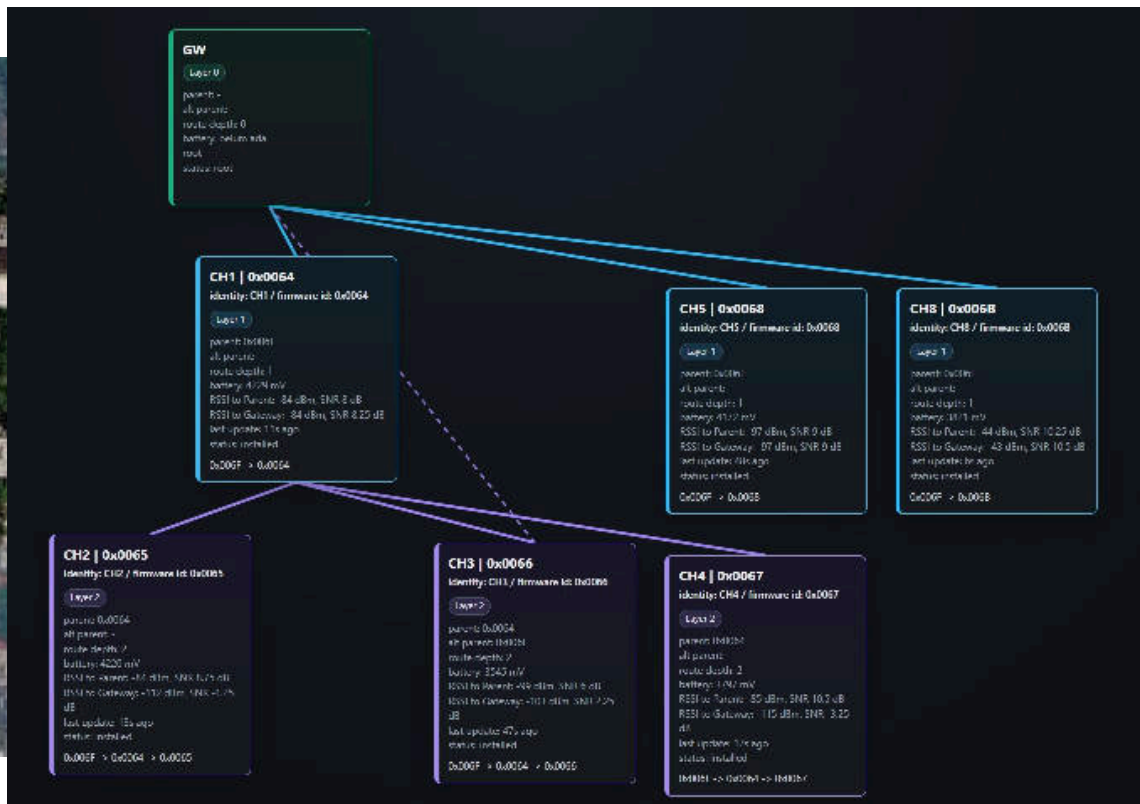
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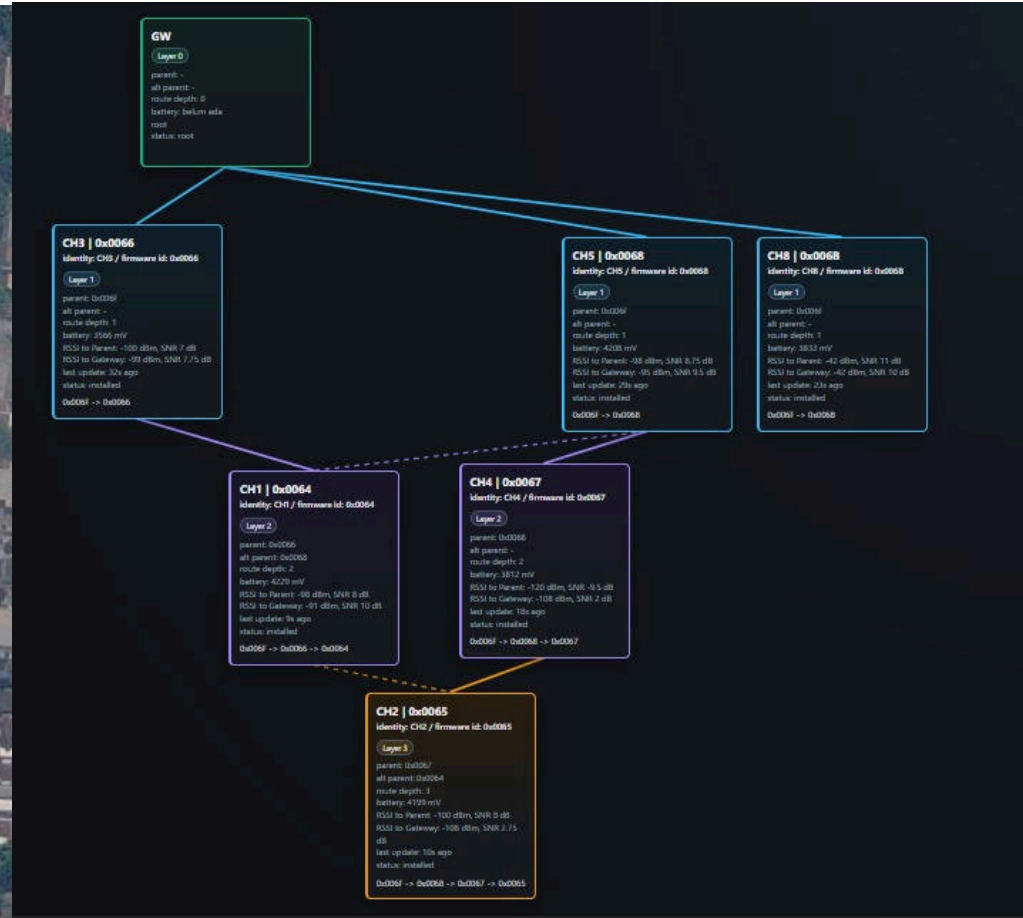
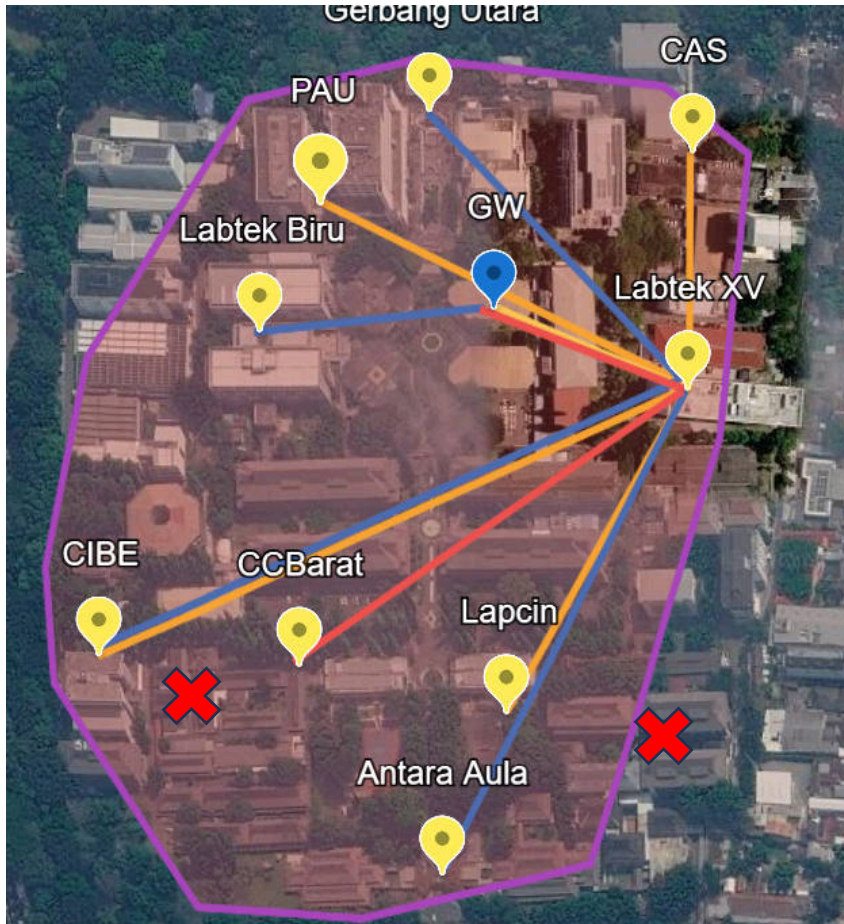




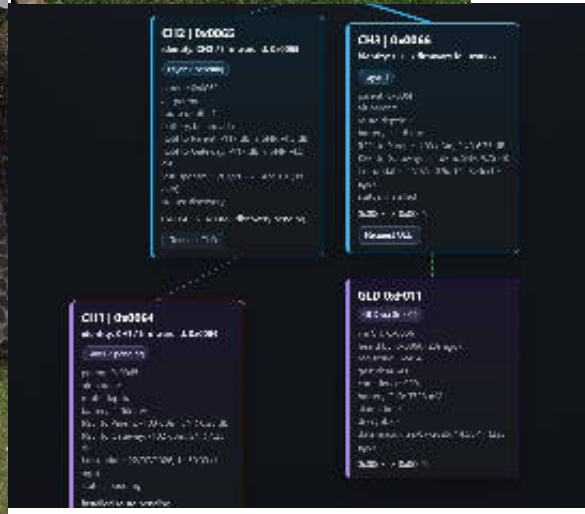
Uji Mesh 2

16 Juli 2026





Uji Downlink GLD 1

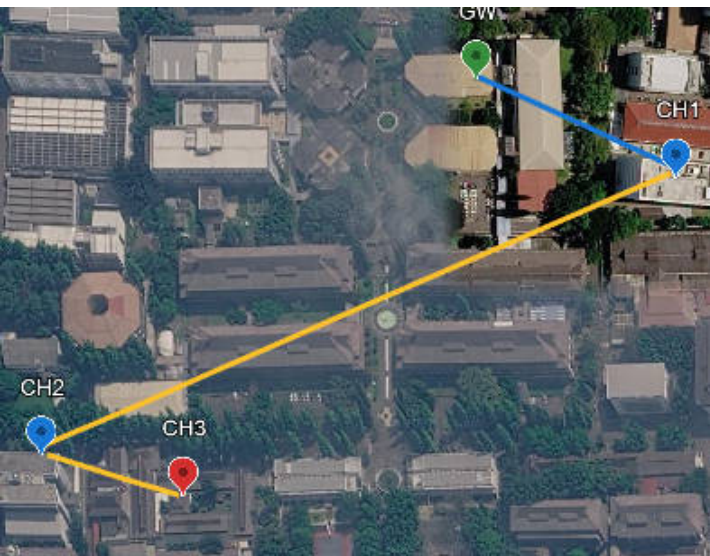


Uji Downlink GLD 2



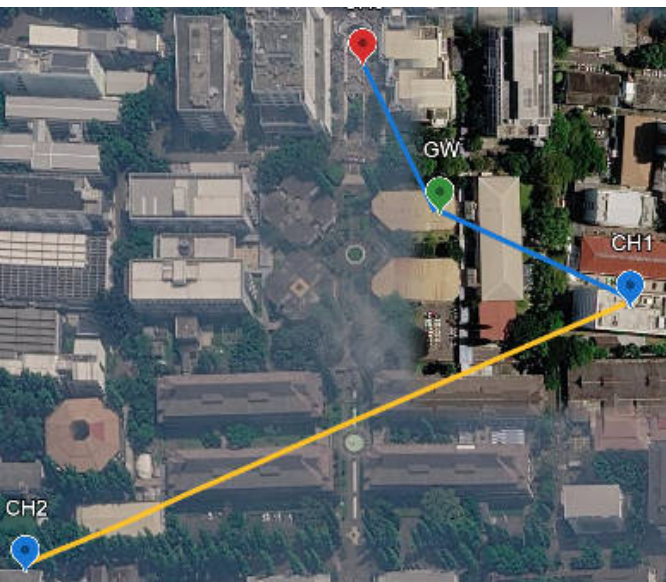
GW Id: 0 Name: GW Type: Gateway Status: Online Signal: 100% Power: 100W
CH1 0x0004 Name: CH1 Channel 1 Id: 1 Name: CH1 Type: Channel Status: Online Signal: 100% Power: 100W
CH2 0x0005 Name: CH2 Channel 2 Id: 2 Name: CH2 Type: Channel Status: Online Signal: 100% Power: 100W
CH3 0x0006 Name: CH3 Channel 3 Id: 3 Name: CH3 Type: Channel Status: Online Signal: 100% Power: 100W

Uji Downlink GLD 3

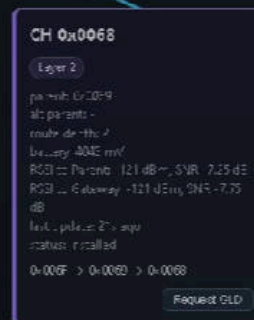


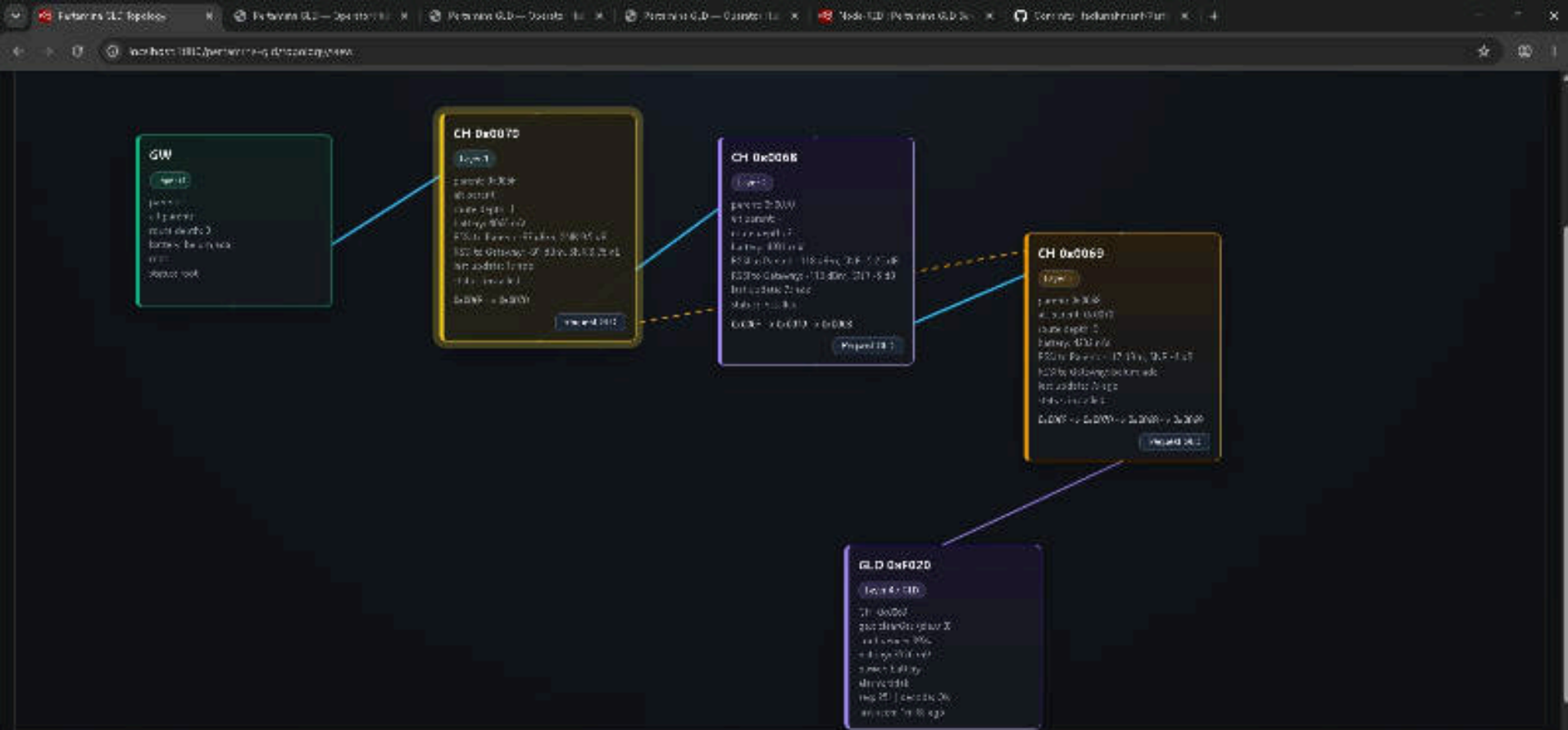
GW [Link] [Data] [Status] [Details]
CH1 0x0004 [Link] [Data] [Status] [Details]
CH2 0x0005 [Link] [Data] [Status] [Details]
CH3 0x0006 [Link] [Data] [Status] [Details]

Uji Downlink GLD 4



Analisis dan Perbaikan Firmware Downlink GLD



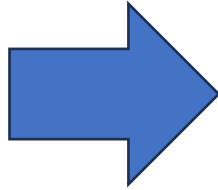


Status CH

Showing 2 CH

CH	Status	Parent	Route	Last CH GLD	Expected Next Event	Role	Update
CH 0x0070	up	0x0000	0x0000 -> 0x0070	11 ago	Received CH GLD packet (battery level: 99%)	Refresh Log	11 ago
CH 0x0069	up	0x0000	0x0000 -> 0x0069	11 ago	Received CH GLD packet (battery level: 99%)	Refresh Log	11 ago

Kebutuhan uji mesh LoRa



Penyangga tiang

Kerjaan Minggu ke-6

01 Uji sistem lingkungan kampus

Pending

02 Evaluasi

Pending

Kerjaan Minggu ke-7

01

Persiapan pemasangan sistem

Pending

List Kerjaan GLD

- 01 Hitung kebutuhan baterai real CH (sekarang 1 baterai 2800 mAh), **penambahan WDT eksternal hardware (pak Maman)**. ❑ CH harus bisa dimanfaatkan untuk berbagai kebutuhan, jadi harus dikulik mendalam
- 02 Hitung kebutuhan panel surya ke CH menggunakan sensor arus (sekarang 6W).
- 03 Mapping area pemasangan CH, utamakan di atas gedung-Gedung ITB dan LoS ❑ antenna fiber
- 04 Optimasi waktu aktivasi node GLD, atur waktu aktif mulai dari 30s, hingga sensor benar-benar stabil. Kemudian Assembly Chamber dan Uji Pembacaan Sensor
- 05 Uji fungsional sistem (GLD-CH-GW-Server), sekarang sedang proses nulling, pengambilan dataset, dan optimasi machine learning
- 06 Uji sistem di lapangan sekitaran ITB, mekanisme pembacaan data sensor dan kirim data ke CH harus disesuaikan, kirim data bisa 10 menit sekali (menghindari traffic tinggi), pembacaan sensor boleh terus menerus

SISTEM GLD

7 Minggu Menuju Sistem Siap Pasang

GLD · CH · CW · Server